CPE334 - Software Engineering

Lab 4: TokTickIT

Actions Taken, dashboards and final regression

Lathapol Srikhiao - 67070503475

Reviewer: Kittakorn Poungpien - 67070503401

GitHub repository

Prepared 6 October 2026

This submission follows the nine answer parts in the Lab 4 handout. Every feature PR was reviewed by Kittakorn-P and merged into lab4-staging; PR #63 released lab4-staging to main. All checks passed on a clean checkout of main commit a4dff3cd228463c841292927c352bdabe69305a5. The reflection is AI-assisted and marked as a draft for the student to rewrite; test execution is attributed to the assistant.

Answer Part 1: Git workflow and review

Answer Part 2: Specification-driven development

Answer Part 3: Tests and traceability

Answer Part 4: AI use and reflection

Answer Part 5: IT Staff dashboard

Answer Part 6: Actions Taken UI

Answer Part 7: Ticket workflow

Answer Part 8: Requester dashboard and regression

Answer Part 9: Zen Green, responsive, accessibility

Answer Part 1

Git use with engineering workflow

Repository source

Lab 4 review record

Author: Lathapol Srikhiao - 67070503475 (@Lathapol). Reviewer: Kittakorn Poungpien - 67070503401 (@Kittakorn-P). Approvals below are taken from the GitHub pull request reviews; each author reply is the comment posted on that PR.

Feature PRs target lab4-staging. The release PR (#63) and the final evidence and PDF PRs (#64, #65) went directly into main. Nothing here was recorded before it existed on GitHub.

Reviewer identity	PR link	Review comment	Author response	Approval
Kittakorn-P	PR #56 Issue 1: sprint contract and test plan	".md look fine the draft is good too."	"ok thanks, i'll start issue 2 now"	Approved and merged into lab4-staging, 2026-09-24
Kittakorn-P	PR #57 Issue 2: Actions Taken foundation	"All the test passed. ready to merge"	"ok thanks, i'll continue issue3 now"	Approved and merged into lab4-staging, 2026-09-24
Kittakorn-P	PR #58 Issue 3: Actions Taken UI	"The code and UI artifact look fine. ready to merge"	"ok thanks, i'll start issue 4 now"	Approved and merged into lab4-staging, 2026-09-24
Kittakorn-P	PR #59 Issue 4: ticket workflow and resolution gate	"All test passed. Ready to merge"	"ok thanks, i'll start issue 5 now"	Approved and merged into lab4-staging, 2026-09-27
Kittakorn-P	PR #60 Issue 5: role dashboards	"All test file passed with valid UI artifacts. Ready to merge"	"ok thanks, i'll start issue 6 now"	Approved and merged into lab4-staging, 2026-09-27
Kittakorn-P	PR #61 Issue 6: final hardening and regression	"All the final & regression complete. Ready to merge"	"ok thanks, i'll open the release pr now"	Approved and merged into lab4-staging, 2026-09-28
Kittakorn-P	PR #62 Issue 7: review record and AI-use record	"I see no problem with this one"	(no reply needed; release PR followed)	Approved and merged into lab4-staging
Kittakorn-P	PR #63 Release: lab4-staging into main	"All change in lab4-staging is ready. merging to main"	"ok thanks, i'll run the final main tests now"	Approved and merged into main, a4dff3c
Kittakorn-P	PR #64 Final main tests and evidence	"all the log and .md is ready"	"ok thanks, i'll finish the pdf now"	Approved and merged into main, 2026-10-13
Kittakorn-P	PR #65 Nine-part submission PDF	"the submission is ready"	"ok thanks, i'll add the project screenshots and my reflection"	Approved and merged into main, 2026-10-13

Review focus: resolution-gate enforcement in the backend, Actions Taken authorization and duplicate/stale handling, additive migration and repeatable seed, backend-calculated dashboards whose counts match their drill-down lists, Lab 1-3 regression, and desktop/tablet/mobile evidence.

Peer review given

The student also reviewed the coworker's own Lab 4 pull requests in Kittakorn-P/toktickit. On PR #49 (Sprint 4 contract) the student commented ".md look fine start on the lab". On PR #50 (Actions Taken foundation) the student's comment listed the issues found (closed and cancelled tickets accepting actions, a 500 on non-string input, a skippable stale check, author-only editing, no double-click protection); the coworker fixed them and the student's second comment recorded that the fixes looked right.

Merge history on main

First-parent history of origin/main shows lab4-staging merged into main by PR #63; the feature PRs were merged into lab4-staging first.

```
66ebdc6 2026-10-13 Merge pull request #65 from Lathapol/feature/7-submission-pdf
5632e90 2026-10-13 Merge pull request #64 from Lathapol/feature/7-final-main-evidence
a4dff3c 2026-10-06 Merge pull request #63 from Lathapol/lab4-staging
8b0a818 2026-09-16 Merge pull request #48 from Lathapol/codex/lab3-final-evidence
9d2c48b 2026-09-29 Merge pull request #62 from Lathapol/feature/7-release-docs
74cb9b2 2026-09-28 Merge pull request #61 from Lathapol/feature/6-final-hardening
d221bba 2026-09-27 Merge pull request #60 from Lathapol/feature/5-role-dashboards
60f3b44 2026-09-27 Merge pull request #59 from Lathapol/feature/4-ticket-workflow
8ed3014 2026-09-24 Merge pull request #58 from Lathapol/feature/3-actions-taken-ui
2c05382 2026-09-24 Merge pull request #57 from Lathapol/feature/2-actions-taken-foundation
0efcabf 2026-09-24 Merge pull request #56 from Lathapol/feature/1-lab4-contract
8b0a818 2026-09-16 Merge pull request #48 from Lathapol/codex/lab3-final-evidence
43e6c06 2026-09-16 Merge pull request #47 from Lathapol/lab3-staging
```

Issues: #49 contract, #50 Actions Taken foundation, #51 Actions Taken UI, #52 workflow, #53 dashboards, #54 hardening, #55 release.

GitHub Project. GITHUB PROJECT SCREENSHOTS: insert the two board screenshots (top and bottom of the board) showing all Lab 4 issues in Done on the next two pages before submitting. They were not captured by the assistant because it cannot read the board.

GitHub Project - top of board (INSERT SCREENSHOT)

GitHub Project - bottom of board (INSERT SCREENSHOT)

Repository setup and structure

README and directory tree

Ignore rules

Environment example

Root: client/ (React UI), server/ (API, Prisma migrations and tests), e2e/ (browser tests), docs/lab-04/ (contract and review records), artifacts/lab-04/ (screenshots, migration report and verification logs), scripts/ (reproducible checks).

Answer Part 2

Specification-driven development

The contract below was written before implementation. Its Definition of Done checkboxes show the pre-implementation state; completion is documented in Parts 1 and 3. It was reviewed and merged as PR #56 on 24 September 2026, before implementation PRs #57 (24 Sep), #58 (24 Sep), #59 (27 Sep), #60 (27 Sep) and #61 (28 Sep) were merged.

Repository source

Sprint 4 engineering contract

Status: written before implementation, from the Lab 4 handout and the merged Lab 3 `main`. Implementation, peer review and final-main test results are recorded elsewhere (`reviewer.md`, `tests.md`); the existence of this file does not imply completion.

1. Sprint goal

Complete the service-desk workflow: IT Staff record Actions Taken under each Ticket, a Ticket cannot be Resolved without documented work, and Requesters, IT Staff and Administrators get concise backend-calculated dashboards. All Lab 1-3 behavior keeps working under the Zen Green design.

2. Stakeholder request (interpretation)

The desk needs a reliable log of the work done on a Ticket. The Ticket Owner still coordinates, but any IT Staff member may log work. Requesters may say "it looks fixed" but only Staff formally resolve. Each role gets a small dashboard whose numbers link to the detailed lists.

3. Scope

Included: Actions Taken (list/create/edit), resolution gate and final transition matrix, Requester/Staff/Admin dashboards, stale-update handling, seed/migration, final hardening (duplicate-submit protection, draft preservation, console/link cleanup, README).

Excluded (handout 4.2): SLA clocks/escalation, on-call, email/SMS/LINE/push, inventory/parts/cost, timesheets/billing, multi-level approval/e-signature, BI/report builders/exports, multi-tenant/cloud operations, and any feature not listed here. Also excluded by decision: deleting Actions Taken, and an "assignee" field on an action (see D-03).

4. Functional requirements

- FR-01 IT Staff and Administrators create an Action Taken under an existing, non-terminal Ticket.

- FR-02 IT Staff and Administrators edit an existing Action Taken.

- FR-03 Ticket Detail lists Actions Taken oldest-first with date/time, description, result, performed-by, follow-up flag/note and attachment notes.

- FR-04 Requesters see all Actions Taken on their own Tickets, read-only; other Requesters' Tickets are 404.

- FR-05 A Ticket moves to RESOLVED only if it has at least one Action Taken.

- FR-06 Status controls show only transitions allowed by the matrix; the backend is the authority.

- FR-07 Requester dashboard: own-ticket metrics and recent lists with drill-down.

- FR-08 Staff dashboard: queue metrics, urgent and recent tickets, current-user actions, drill-down. Administrator gets the Staff dashboard plus user-account counts.

- FR-09 Dashboard navigation per role with active-page indication and loading/empty/forbidden/failure states.

- FR-10 Repeated clicks/retries do not create duplicate Actions Taken; failed forms keep their entered data.

- FR-11 Lab 1-3 features (auth, create/list/detail/attachments, queue, ownership, comments/notes, user management) remain unchanged and tested.

5. Business rules

- BR-01 An Action Taken belongs to exactly one Ticket; a Ticket has many.

- BR-02 The Ticket Owner coordinates, but any active IT Staff/Administrator may create or edit an Action on any Ticket, regardless of who owns it.

- BR-03 **Performed by** is set by the server to the authenticated creator; clients cannot supply or change it. Editing by another Staff member does not change it.

- BR-04 **Action date/time** is server-generated at creation (`createdAt`), never client-supplied, never edited. Actions list oldest-first, ties by id.

- BR-05 Description: trimmed, 1-2000 chars. Result: trimmed, 1-2000 chars. Attachment Notes: optional, trimmed, ≤500 chars (free text naming which file/image to look for; no upload).

- BR-06 `followUpRequired` is boolean (default false). When true, `followUpNote` is required, trimmed, 1-1000 chars. When false, the note is stored as null (a supplied note is discarded).

- BR-07 Actions can be created/edited only while the Ticket is NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED or REOPENED. CLOSED and CANCELLED Tickets are read-only for Actions (409).

- BR-08 Actions are never deleted (audit trail). Edits increment the action `version`; an update with a stale version returns 409.

- BR-09 Only IT Staff and Administrator write Actions. Requesters never write; all writes are enforced by the backend, not hidden controls.

- BR-10 **Resolution gate:** changing a Ticket to RESOLVED requires >=1 Action Taken on that Ticket, else 409 `ACTION_REQUIRED`. Checked in the same transaction as the status update, so a client bypassing the UI is still blocked. Legacy Tickets already RESOLVED/CLOSED are not changed by migration; they simply show zero actions.

- BR-11 Requester "appears resolved" remains advisory; it never changes `currentStatus` (Lab 3 BR-05 preserved).

- BR-12 Status transitions use the matrix in §5.2. Terminal states CLOSED/CANCELLED have no exits. Confirmation is required for RESOLVED, CLOSED, CANCELLED. Only IT Staff change status (Admin read-only, as in Lab 3).

- BR-13 Ticket updates and Action updates carry a version; stale writes return 409 and never overwrite. Creating an Action also touches the Ticket `updatedAt` (Ticket version unchanged) so "recently updated" reflects work.

- BR-14 Duplicate-create protection: `POST` requires a client-generated `requestKey` (UUID, unique per Ticket). A repeat with the same key returns the original Action with 200 instead of creating a second one.

- BR-15 **Dashboards** are calculated by the backend from current database rows for the authenticated user, never from client input. "Open" group = NEW, OPEN, IN_PROGRESS, WAITING_FOR_REQUESTER, REOPENED. "Recent" = `updatedAt` >= now − 7×24 h computed by the server; lists show the 5 newest,

ordered `updatedAt` desc then id desc. Date boundaries are UTC instants; display uses the browser locale (Asia/Bangkok in the lab).

- BR-16 Requester dashboard sees only `requesterId = current user`. Staff dashboard covers all Tickets. Requesters get 403 on the staff dashboard; Staff/Admin get 403 on the requester dashboard.

- BR-17 Every dashboard count has a drill-down target whose list uses the same filter, so the count equals the list `totalCount`. Zero results return `0` and empty lists, never an error.

- BR-18 Errors are safe and structured `{error:{code,message}}` (400 validation, 401, 403, 404 missing/foreign, 409 conflict/gate/terminal, 500 generic). No stack traces, SQL, paths or notes.

- BR-19 Lab 3 rules BR-01..BR-16 (auth, CSRF, privacy of Internal Notes, attachments, ownership) still apply unchanged.

5.1 Authorization matrix (adds to Lab 3)

Operation	Requester	IT Staff	Administrator
List Actions on a Ticket	Own Tickets	All	All
Create / edit Action	Denied (403)	Allowed	Allowed
Requester dashboard	Own	Denied	Denied
Staff dashboard	Denied	Allowed	Allowed (+ user counts)
Ticket status change	Denied	Allowed	Denied (Lab 3)

5.2 Final status transition matrix

Destinations unchanged from Lab 3, with the gate added. Authorized role: IT Staff only.

From	Allowed destinations	Extra condition
NEW	OPEN, CANCELLED	
OPEN	IN_PROGRESS, WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	RESOLVED needs >=1 Action
IN_PROGRESS	WAITING_FOR_REQUESTER, RESOLVED, CANCELLED	RESOLVED needs >=1 Action
WAITING_FOR_REQUESTER	IN_PROGRESS, RESOLVED, CANCELLED	RESOLVED needs >=1 Action
RESOLVED	CLOSED, REOPENED	
REOPENED	OPEN, IN_PROGRESS, CANCELLED	
CLOSED / CANCELLED	none	terminal

5.3 Dashboard metrics

Card	Definition (server query)	Empty	Drill-down
R: Open tickets	own, status in Open group	0	`/tickets?group=open`
R: Waiting for you	own, WAITING_FOR_REQUESTER	0	`/tickets?status=WAITING_FOR_REQUESTER`
R: Recently updated	own, updatedAt >= now−7d, top 5	"No recent updates"	each row -> Ticket Detail
R: Recently resolved	own, RESOLVED or CLOSED, updatedAt >= now−7d, top 5	"Nothing resolved recently"	each row -> Ticket Detail; count -> `/tickets?group=resolved&recent=7d`
S: Unassigned	ownerId null, Open group	0	queue `owner=unassigned&group=open`
S: My tickets	ownerId = me, Open group	0	queue `owner=mine&group=open`

Card	Definition (server query)	Empty	Drill-down
S: By status	count per each of the 8 statuses	0 each	queue `status=X`
S: By IT priority	count per LOW/MEDIUM/HIGH, Open group	0 each	queue `priority=X&group=open`
S: Urgent	itPriority HIGH, Open group, top 5, oldest `updatedAt` first	"No urgent tickets"	row -> Detail
S: Recently updated	any owner, updatedAt >= now−7d, top 5	"No recent updates"	row -> Detail
S: My recent actions	Actions where performer = me, last 7d: count + 5 newest	0 / "No actions logged"	row -> Ticket Detail
A: User accounts	active count per role, inactive count	0	`/users`

6. UI specification summary

See `ui-spec.md`. Ticket Detail (staff) gains an Actions Taken panel (list, create form, edit form). Requester Detail gains the same list read-only. A Dashboard page per role becomes the role home; navigation shows the active page. All screens reuse Zen Green tokens, badges and feedback components.

7. Data changes

New model `ActionTaken` (table `ActionTaken`): `id` PK, `ticketId` FK -> Ticket, `performedById` FK -> User, `description`, `result`, `followUpRequired` bool default false, `followUpNote` null, `attachmentNotes` null, `requestKey` text, `version` int default 0, `createdAt`, `updatedAt`. Indexes: `(ticketId, createdAt, id)` for the list and resolution gate, `(performedById, createdAt)` for "my recent actions", unique `(ticketId, requestKey)` for BR-14. Add `Ticket(updatedAt)` for recent lists and `Ticket(ownerId, currentStatus)` for owned counts.

Migration: additive SQL only (CREATE TABLE + CREATE INDEX). No existing table or row changes.

Backfill: none needed. Legacy Tickets have zero Actions: detail shows "No actions recorded", dashboards count them normally, and Tickets already RESOLVED are unaffected (the gate applies only to new transitions).

Recovery: reversible with `DROP TABLE "ActionTaken"`. Test by migrating a restored Lab 3 backup, comparing row counts/digests of all Lab 3 tables before and after, then dropping the table and re-comparing.

Seed (idempotent): insert only if absent, keyed by fixed ticket numbers and requestKeys. Tickets cover all 8 statuses, LOW/MEDIUM/HIGH, owned and unassigned, with 0, 1 and 3+ Actions (multiple performers on one Ticket), a follow-up example, and one Requester with zero tickets to demonstrate zero dashboard metrics. Re-running changes nothing.

8. API contract

See `api-spec.md`.

9. Acceptance criteria

ID	Observable outcome
AC-01	Permitted Staff + valid data creates an Action under the right Ticket with the authenticated performer and server timestamp.
AC-02	Validation: empty/over-length description or result, follow-up true without note, bad JSON/unknown fields all return 400 with safe messages.

ID	Observable outcome
AC-03	Staff can edit an Action; performer and date do not change; a stale version returns 409 and changes nothing.
AC-04	A Staff member who is not the Ticket Owner can log an Action (BR-02); performer is that member.
AC-05	Requester cannot create/edit (403); Requester reads Actions only on own Ticket (foreign -> 404).
AC-06	Actions on CLOSED/CANCELLED Tickets are rejected 409; on other states accepted.
AC-07	Repeating a create with the same requestKey yields one row, including concurrent repeats.
AC-08	Status -> RESOLVED with zero Actions returns 409 `ACTION_REQUIRED` even via direct API; with >=1 Action it succeeds.
AC-09	Every matrix transition works for Staff only; invalid, terminal, unconfirmed and stale updates fail; Requester appears-resolved does not change status.
AC-10	Requester dashboard returns only the caller's metrics/lists; another Requester's Tickets never appear.
AC-11	Staff dashboard metrics equal direct database counts; Admin also gets user counts; Requester/unauthenticated denied.
AC-12	Each dashboard count equals the `totalCount` of its drill-down list.
AC-13	Empty datasets return zeros/empty lists and the UI shows empty states.
AC-14	Ticket Detail shows Actions list, create mode, edit mode and a read-only view for Requesters; the summary status refreshes after a status change.
AC-15	Dashboard UIs show loading, error+retry, forbidden, empty and populated states and are role-navigated.
AC-16	Migration preserves all Lab 3 data; seed is repeatable; legacy Tickets show zero actions.
AC-17	Desktop/tablet/mobile layouts have visible focus, labels, no clipping or horizontal overflow.
AC-18	Full Lab 1-3 regression (server, client, E2E) passes on final `main`.

10. Definition of Done

- [ ] AC-01..AC-18 each map to a passing automated test (see `tests.md`).

- [ ] Additive migration applied to a restored Lab 3 backup with before/after evidence; seed run twice with identical results.

- [ ] Server, client and Playwright suites pass on `main`; TypeScript and Vite builds pass.

- [ ] No console errors, broken links, placeholder text or unfinished controls in any role's screens.

- [ ] Desktop/tablet/mobile screenshots for Staff dashboard, Requester dashboard and Actions Taken, inspected against the ui-spec checklist.

- [ ] Dashboard sample metrics verified against direct database queries.

- [ ] README (setup, migrate, seed, test, demo) current; `reviewer.md` and `ai-use.md` truthful.

- [ ] Feature PRs reviewed into `lab4-staging`, then `main`; all Issues Done.

11. Assumptions and decisions

- D-01 Resolution gate = at least one Action Taken (simple, testable, backend-enforced); it does not require follow-ups to be cleared.

- D-02 **DB decision:** a separate `ActionTaken` table (not JSON on Ticket) so many-per-ticket, per-row audit, indexes and joins for dashboards work.

- D-03 **DB decision:** no delete and no client timestamp; `createdAt` doubles as "Action date/time" so records cannot be backdated. `requestKey` with a unique index gives idempotency at the database level, not just in code. No assignee column: the handout lists "Performed by (auto)"; assignment stays on Ticket owner, and inactive-owner rejection remains the Lab 3 rule.

- D-04 Optimistic concurrency via integer `version`, matching Lab 3.

- D-05 Requesters see all Actions on their own Tickets (handout 8.3 resolves the "where approved" wording in 4.3). Actions hold no private data, unlike Internal Notes.

- D-06 "Recent" is 7 days rolling, top 5 rows, to keep dashboards concise.

- D-07 `group=open` and `group=resolved` are small additive list filters so dashboard counts and drill-downs match.

Answer Part 3

Test DD and traceability

Repository source

Lab 4 test plan and traceability

Written alongside the specification and before implementation. Final status is filled in as each issue merges; Every row is now Pass; see the integrated results below. Tests use the isolated `toktickit_lab4_test` database (the Jest guard refuses other names).

Test ID	Type	AC	What it tests	Expected	Automated test file	Final
UNIT-01	Unit	AC-02	Action validation (lengths, follow-up rule, unknown keys)	Rejects invalid, trims valid	server/tests/lab-04/actions-validation.test.ts	Pass
UNIT-02	Unit	AC-08,09	Transition matrix + gate helper	Every from/to pair per §5.2	server/tests/lab-04/ticket-workflow.api.test.ts	Pass
API-01	API	AC-01	Create Action	201, correct ticket, performer, server date	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-02	API	AC-02	Invalid bodies	400 safe errors	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-03	API	AC-03	Edit + stale version	200 then 409, no change	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-04	API	AC-04	Non-owner staff logs action	performer = that staff	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-05	API	AC-05	Requester write denied, foreign read 404	403 / 404	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-06	API	AC-06	Terminal Ticket rejects actions	409 TICKET_TERMINAL	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-07	API	AC-07	Duplicate requestKey, concurrent	One row	server/tests/lab-04/actions-taken.api.test.ts	Pass
API-08	API	AC-08	Resolve without / with action	409 ACTION_REQUIRED / 200	server/tests/lab-04/ticket-workflow.api.test.ts	Pass
API-09	API	AC-09	All transitions, roles, unconfirmed, stale, advisory signal	Per matrix	server/tests/lab-04/ticket-workflow.api.test.ts	Pass
API-10	API	AC-10	Requester dashboard isolation	Only own data	server/tests/lab-04/requester-dashboard.api.test.ts	Pass

Test ID	Type	AC	What it tests	Expected	Automated test file	Final
API-11	API	AC-11	Staff/Admin dashboard vs direct DB counts; role denials	Equal; 403/401	server/tests/lab-04/staff-dashboard.api.test.ts	Pass
API-12	API	AC-12	Count equals drill-down totalCount	Equal	server/tests/lab-04/staff-dashboard.api.test.ts; server/tests/lab-04/requester-dashboard.api.test.ts	Pass
API-13	API	AC-13	Empty dataset	zeros/empty lists	server/tests/lab-04/requester-dashboard.api.test.ts	Pass
MIG-01	Migration	AC-16	Lab 3 backup preserved, seed twice identical, rollback	Digests match	scripts/verify-lab4-migration.cjs; server/tests/lab-04/seed.api.test.ts	Pass
COMP-01	UI component	AC-14	Actions list, create/edit mode, read-only for Requester, validation, draft kept	Pass	client/tests/lab-04/ActionsTaken.test.tsx	Pass
COMP-02	UI component	AC-08,14	Status controls show allowed transitions, gate hint, confirm dialog	Pass	client/tests/lab-04/TicketWorkflow.test.tsx	Pass
COMP-03	UI component	AC-15	Staff dashboard states + links	Pass	client/tests/lab-04/StaffDashboard.test.tsx	Pass
COMP-04	UI component	AC-10,15	Requester dashboard states + links	Pass	client/tests/lab-04/RequesterDashboard.test.tsx	Pass
E2E-01	E2E	AC-01,03,04,14	Staff creates/edits actions on a ticket, multiple performers	Pass	e2e/lab-04/actions-taken-flow.spec.ts; e2e/lab-04/accessibility.spec.ts	Pass
E2E-02	E2E	AC-08,09	Resolution gate and close flow	Pass	e2e/lab-04/ticket-resolution.spec.ts	Pass
E2E-03	E2E	AC-11,12,15	Dashboards + drill-down, all roles	Pass	e2e/lab-04/dashboards.spec.ts	Pass
RESP-01	UI style/responsive	AC-17	Screens at 1440/820/390: no overflow/clipping, focus visible	Pass	e2e/lab-04/dashboards.spec.ts; e2e/lab-04/actions-taken-flow.spec.ts; e2e/lab-04/accessibility.spec.ts	Pass
A11Y-01	Accessibility	AC-17	Labels, aria-current, dialog focus, keyboard-only flow	Pass	e2e/lab-04/actions-taken-flow.spec.ts; e2e/lab-04/accessibility.spec.ts	Pass

Test ID	Type	AC	What it tests	Expected	Automated test file	Final
PERF-01	Performance smoke	AC-11	Staff dashboard on seeded data responds under 1 s	Pass	server/tests/lab-04/staff-dashboard.api.test.ts	Pass
REG-01	Regression	AC-18	Labs 1-3 server, client and E2E suites	All pass	server/tests/lab-01..03; client/tests/lab-01..03; e2e/lab-02..03	Pass

Traceability rule: each of AC-01..AC-18 appears in at least one row above (AC-01..18 covered by rows API-01..13, MIG-01, COMP-01..04, E2E-01..03, RESP-01, A11Y-01, REG-01).

Integrated results (issue 6)

`node scripts/verify-lab4.cjs` against `toktickit_lab4_test`: server 19 suites / 261 tests, client 11 files / 59 tests, server and client type checks, Vite build, migration check (Lab 3 data preserved, repeat seed identical, rollback restores Lab 3), Lab 4 browser suite 12 scenarios and Lab 3 browser regression 15 scenarios (desktop 1440x1000, tablet 820x1180, mobile 390x844) all pass. Commands, exit codes and logs are in `artifacts/lab-04/verification/` (report.json). Final-main verification is recorded in the section below.

Final-main verification (issue 7)

Tested commit: `a4dff3cd228463c841292927c352bdabe69305a5` (PR #63, lab4-staging merged into main), run on 2026-10-06 from a clean checkout created directly from origin/main with `node scripts/verify-lab4.cjs` against `toktickit_lab4_test`. All eight steps exited 0: server 19 suites / 261 tests; client 11 files / 59 tests; server and client type checks; Vite build; migration check (Lab 3 data preserved, repeat seed identical, rollback restores Lab 3); Lab 4 browser suite 12 scenarios; Lab 3 browser regression 15 scenarios, at 1440x1000, 820x1180 and 390x844. Exact commands, exit codes and logs: report.json and the sibling .log files. Every row in the traceability table above is Pass and every AC-01..AC-18 is covered. Later edits to evidence documents do not change the application.

Final-main results

Passed on clean main commit a4dff3cd228463c841292927c352bdabe69305a5 on 6 October 2026. The complete terminal outputs follow; ANSI styling is removed and long lines wrap for readability.

Tested commit recorded in report.json: a4dff3cd228463c841292927c352bdabe69305a5

server-tests full output - exit 0

server-build full output - exit 0

client-tests full output - exit 0

client-types full output - exit 0

client-build full output - exit 0

migration-check full output - exit 0

browser-tests-lab4 full output - exit 0

browser-tests-lab3-regression full output - exit 0

server-tests - complete output

```
(node:7804) DeprecationWarning: Calling client.query() when the client is already executing a query is
```

```
deprecated and will be removed in pg@9.0. Use async/await or an external async flow control mechanism
instead.
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:7804) DeprecationWarning: Calling client.query() when the client is already executing a query is
deprecated and will be removed in pg@9.0. Use async/await or an external async flow control mechanism
instead.

Test Suites: 19 passed, 19 total
Tests:       261 passed, 261 total
Snapshots:   0 total
Time:        58.897 s
Ran all test suites.
```

server-build - complete output

```
(No terminal output; exit code 0.)
```

client-tests - complete output

```
 RUN  v4.1.10 C:/Users/User/AppData/Local/Temp/main-verify/client


 Test Files  11 passed (11)
      Tests  59 passed (59)
   Start at  19:25:29
   Duration  39.04s (transform 13.95s, setup 62.30s, import 16.87s, tests 9.03s, environment 311.13s)
```

client-types - complete output

```
(No terminal output; exit code 0.)
```

client-build - complete output

```
vite v8.2.1 building client environment for production...

transforming...PASS 32 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.46 kB ? gzip:  0.30 kB
dist/assets/index-BLS7npLo.css  240.35 kB ? gzip: 33.58 kB
dist/assets/index-C34Aklwk.js   259.89 kB ? gzip: 75.10 kB

PASS built in 1.63s
```

migration-check - complete output


```
{
  "checkedAt": "2026-10-06T12:26:30.516Z",
  "sourceTemplate": "toktickit_lab3_test",
  "evidenceDatabase": "toktickit_lab4_evidence_1791289576896",
  "lab3Counts": {
    "RequesterUser": 26,
    "Ticket": 164,
    "Attachment": 38,
    "Category": 4,
    "RelatedSystem": 7,
    "TicketEntry": 48,
    "Session": 1
  },
  "lab3DigestBefore": "5edc5126cf0e51576114bbfb43bb2c7c6cf7f48a20070790de9efa692357b21f",
  "lab3DigestAfterMigration": "5edc5126cf0e51576114bbfb43bb2c7c6cf7f48a20070790de9efa692357b21f",
  "lab3RowsPreserved": true,
  "legacyActionsCreatedByMigration": 0,
  "afterSeedCounts": {
    "RequesterUser": 27,
    "Ticket": 164,
    "Attachment": 38,
    "Category": 4,
    "RelatedSystem": 7,
    "TicketEntry": 48,
    "ActionTaken": 24
  },
  "ticketsWithZeroOneManyActions": {
    "zero": 152,
    "one": 6,
    "many": 6
  },
  "firstSeedDigest": "6377e4780c093a3bb1b6de3457ab7834abfd364d1d1e6ef464727149a9cc8d58",
  "secondSeedDigest": "6377e4780c093a3bb1b6de3457ab7834abfd364d1d1e6ef464727149a9cc8d58",
```

```
  "repeatSeedUnchanged": true,
  "rollbackRestoredLab3": true,
  "rollbackDigest": "5edc5126cf0e51576114bbfb43bb2c7c6cf7f48a20070790de9efa692357b21f"
}
```

browser-tests-lab4 - complete output

```
Running 12 tests using 1 worker

  ok  1 [desktop] > e2e\lab-04\accessibility.spec.ts:11:5 > A11Y-01 keyboard-only flow, visible focus,
landmarks and a clean console for every role (9.1s)
  ok  2 [desktop] > e2e\lab-04\actions-taken-flow.spec.ts:10:5 > E2E-01 staff add and edit actions,
different performers, requester read-only, responsive (9.2s)
  ok  3 [desktop] > e2e\lab-04\dashboards.spec.ts:11:5 > E2E-03 role dashboards show metrics, drill-down to
the filtered list, and rows open a ticket (3.2s)
  ok  4 [desktop] > e2e\lab-04\ticket-resolution.spec.ts:11:5 > E2E-02 resolution gate: cannot resolve
without an action, resolves after one, requester signal stays advisory (6.1s)
  ok  5 [mobile] > e2e\lab-04\accessibility.spec.ts:11:5 > A11Y-01 keyboard-only flow, visible focus,
landmarks and a clean console for every role (3.9s)
  ok  6 [mobile] > e2e\lab-04\actions-taken-flow.spec.ts:10:5 > E2E-01 staff add and edit actions, different
 performers, requester read-only, responsive (6.0s)
  ok  7 [mobile] > e2e\lab-04\dashboards.spec.ts:11:5 > E2E-03 role dashboards show metrics, drill-down to
the filtered list, and rows open a ticket (2.4s)
  ok  8 [mobile] > e2e\lab-04\ticket-resolution.spec.ts:11:5 > E2E-02 resolution gate: cannot resolve
without an action, resolves after one, requester signal stays advisory (4.9s)
  ok  9 [tablet] > e2e\lab-04\accessibility.spec.ts:11:5 > A11Y-01 keyboard-only flow, visible focus,
landmarks and a clean console for every role (5.5s)
  ok 10 [tablet] > e2e\lab-04\actions-taken-flow.spec.ts:10:5 > E2E-01 staff add and edit actions, different
 performers, requester read-only, responsive (9.5s)
  ok 11 [tablet] > e2e\lab-04\dashboards.spec.ts:11:5 > E2E-03 role dashboards show metrics, drill-down to
the filtered list, and rows open a ticket (2.6s)
  ok 12 [tablet] > e2e\lab-04\ticket-resolution.spec.ts:11:5 > E2E-02 resolution gate: cannot resolve
without an action, resolves after one, requester signal stays advisory (6.6s)

  12 passed (1.2m)
(node:10728) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
(node:28200) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
(node:25732) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
```

browser-tests-lab3-regression - complete output

```
Running 15 tests using 1 worker

  ok  1 [desktop] > e2e\lab-03\queue.spec.ts:10:6 > IT_STAFF queue filters, pagination and read-only detail
(2.6s)
  ok  2 [desktop] > e2e\lab-03\queue.spec.ts:10:6 > ADMINISTRATOR queue filters, pagination and read-only
detail (1.8s)
  ok  3 [desktop] > e2e\lab-03\requester.spec.ts:18:5 > login, first change, create with attachment, filter,
 reload, remove and logout (3.6s)
  ok  4 [desktop] > e2e\lab-03\users.spec.ts:8:5 > admin creates, edits, deactivates and resets an account;
user must change password (5.3s)
  ok  5 [desktop] > e2e\lab-03\workflow.spec.ts:10:5 > staff workflow, private notes, requester signal and
admin read-only view (13.1s)
  ok  6 [mobile] > e2e\lab-03\queue.spec.ts:10:6 > IT_STAFF queue filters, pagination and read-only detail
(10.1s)
  ok  7 [mobile] > e2e\lab-03\queue.spec.ts:10:6 > ADMINISTRATOR queue filters, pagination and read-only
detail (3.9s)
  ok  8 [mobile] > e2e\lab-03\requester.spec.ts:18:5 > login, first change, create with attachment, filter,
reload, remove and logout (6.3s)
  ok  9 [mobile] > e2e\lab-03\users.spec.ts:8:5 > admin creates, edits, deactivates and resets an account;
user must change password (8.6s)
  ok 10 [mobile] > e2e\lab-03\workflow.spec.ts:10:5 > staff workflow, private notes, requester signal and
admin read-only view (14.4s)
  ok 11 [tablet] > e2e\lab-03\queue.spec.ts:10:6 > IT_STAFF queue filters, pagination and read-only detail
(2.2s)
  ok 12 [tablet] > e2e\lab-03\queue.spec.ts:10:6 > ADMINISTRATOR queue filters, pagination and read-only
detail (2.5s)
  ok 13 [tablet] > e2e\lab-03\requester.spec.ts:18:5 > login, first change, create with attachment, filter,
reload, remove and logout (5.0s)
  ok 14 [tablet] > e2e\lab-03\users.spec.ts:8:5 > admin creates, edits, deactivates and resets an account;
user must change password (6.6s)
  ok 15 [tablet] > e2e\lab-03\workflow.spec.ts:10:5 > staff workflow, private notes, requester signal and
admin read-only view (9.2s)

  15 passed (1.7m)
(node:23172) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
(node:20832) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
```

```
(Use `node --trace-warnings ...` to show where the warning was created)
(node:8904) Warning: The 'NO_COLOR' env is ignored due to the 'FORCE_COLOR' env being set.
(Use `node --trace-warnings ...` to show where the warning was created)
```

Answer Part 4

AI use with reflection

Repository source

Lab 4 AI-use record

Assistant: Claude (model Claude Sonnet 5) used through Claude Code in the Claude desktop app, for both the specification (contract) work and the coding work. The specification was written by the assistant from the Lab 4 handout and reviewed by the coworker in PR #56 before any implementation PR was merged.

Selected genuine user prompts (actual messages from the working conversation; typos kept):

1. Attached SE Lab 4.pdf and the repository link with: "this is for my ticktokIT for my software engi work"

2. Answered the assistant's setup question with the local repository ("It's already local") and: "walk me through tell me what i need to do and what 're we're doing e=for each issues then you do it and told me when finish"

3. Attached the Lab 3 submission PDF with: "go and this is my lab3 for ref"

4. Attached a screenshot of the GitHub Project board with: "also do these and you can automatically open a pull req for me"

5. "also look at my prvious format when creating issues make the name easy to read like mine too and answering comment and commented on my peer pullreq liek what i do too copy my style"

6. "dont forhget to do the broad too"

7. "chekc my freind pull req", followed by "yeah tell him that" and "ok he said he fixed it"

8. "both my issuse 1 and 2 are merge proceed on issues 3"

9. "let do issues 4 now", "proceed on with issues 5", "contionue with issuse 6"

10. "how much issues we have left", then "let go on issues 7"

Work by issue (evidence is in each issue-0X.md; approvals are in reviewer.md):

- Issue 1: the assistant read the handout and the Lab 3 repository, then drafted the engineering contract, API and UI specifications and the test plan before any code. Two database decisions (a separate ActionTaken table; no delete and no client timestamps, with an idempotency key) are justified in specification.md.

- Issue 2: additive migration, idempotent seed and the Actions Taken API with tests. A reviewer-facing note: a first attempt at keeping the repository's mixed line endings produced a noisy diff, which was redone so the pull request shows only the real changes.

- Issue 3: Actions Taken screen with create/edit, read-only requester view and responsive layout, checked with component and browser tests at three viewports.

- Issue 4: the resolution gate (a ticket cannot become Resolved without an action), checked under the ticket lock. This changed Lab 3 behavior, so the Lab 3 fixtures and browser flow were updated.

- Issue 5: backend-calculated dashboards whose counts equal their drill-down lists. Landing on a dashboard changed the first screen for every role, so eleven existing browser tests were updated.

- Issue 6: removed leftover Lab 2 code, restored a visible keyboard focus ring that Bootstrap was hiding (found by the new accessibility browser test), added the integrated verification script and updated the README.

- The student also reviewed the coworker's Lab 4 pull requests; see reviewer.md.

No test result, approval or reflection in this file was invented. Prompts written by the assistant are not attributed to the student.

My Reflection

> DRAFT written by the assistant for the student to read, correct and rewrite in their own words before submission. It is not evidence that the student personally ran the assistant's tests.

The biggest lesson of this lab is that a rule such as "a ticket cannot be resolved without documented work" only counts if the backend enforces it. The screen already hides the option, but the real check has to live in the same transaction as the status change, otherwise a direct request or two staff acting at once can bypass it. The same idea applied to Actions Taken: the performer and time come from the server, a repeat click cannot create a duplicate, and an out-of-date edit is refused instead of silently overwriting someone else.

Writing the contract first made the later issues easier to review, because decisions such as the seven-day "recent" window, the open-ticket group and where each dashboard card links were settled once. Making every dashboard number match the list it opens caught a real gap early. The browser tests also found problems that API tests could not, such as a hidden keyboard focus ring, and a change to the landing page that quietly broke eleven older tests.

Working with an AI assistant was fastest when I gave short instructions and checked the results myself: reading the screenshots, asking my coworker to review, and reviewing my coworker's pull request and asking for fixes. Next time I would run the full regression earlier and keep the pull requests small, since the review comments were most useful when each pull request did one thing.

Answer Part 5

Working IT Staff dashboard UI

Metrics are calculated by the backend (see API-11, API-12 and PERF-01 in Part 3): dashboard counts are compared with direct database counts and with the totalCount of the matching drill-down list, role denials return 403/401, and empty datasets return zeros. The browser test dashboards.spec.ts clicks a card into the filtered Ticket Queue, clears the filter banner and opens a ticket from a list row. Component tests cover loading, empty, forbidden and safe-failure states. The numbers shown are from the shared local test database, which holds many fixture tickets.

Repository source

lab4 issue5 : Role dashboards

Depends on #50 (Actions Taken API, PR #57) and #52 (workflow/resolution gate, PR #59). Adds the Requester and Staff/Admin dashboards.

What changed

- **API:** `GET /api/dashboard/requester` (Requester only) and `GET /api/dashboard/staff` (Staff/Admin, Administrator also gets `users` counts). Both computed by the backend in one `RepeatableRead` transaction from live rows, never client input. Definitions match specification.md §5.3: open group = NEW/OPEN/IN_PROGRESS/WAITING_FOR_REQUESTER/REOPENED, "recent" = last 7 days, lists cap at 5.

- **Drill-down parity:** `GET /tickets` and `GET /staff/tickets` gained `group=open|resolved` and `recent=7d` filters (intersecting with `status`/`priority`/etc.), so every dashboard count equals its drill-down list's `totalCount` (BR-17).

- **UI:** new `RequesterDashboard` and `StaffDashboard` pages, both the new role home. Cards for the counts, breakdown lists for status/IT priority, row lists for urgent/recently updated/recently resolved/my recent actions, each row and card clickable. `MyTickets` and `TicketQueue` accept an `initialFilter` and show a dismissible "Showing ... from the dashboard" banner when arriving via drill-down.

- **Navigation:** every role's shell now shows **Dashboard** first, with `aria-current="page"` on the active link; it's the landing page after login for all three roles.

- Existing Lab 1-3 e2e specs and the Lab 3 `Auth.test.tsx` fixture updated for the new Dashboard landing page (they now navigate to My Tickets/Ticket Queue/Users explicitly before doing their own checks).

Verification - 2026-09-27

- Backend: 19 suites, 261 tests passed against `toktickit_lab4_test`, including dashboard metrics checked against direct database counts, role denials, own-data isolation, empty states, and count-equals-drill-down-list assertions (AC-10..13). A performance-smoke test confirms the staff dashboard responds under 1s on seeded data.

- Client: 11 files, 58 tests passed (14 new in `StaffDashboard.test.tsx`/`RequesterDashboard.test.tsx`), covering loading/error/forbidden/empty states and every card/row's drill-down target. `tsc -b` and Vite build passed.

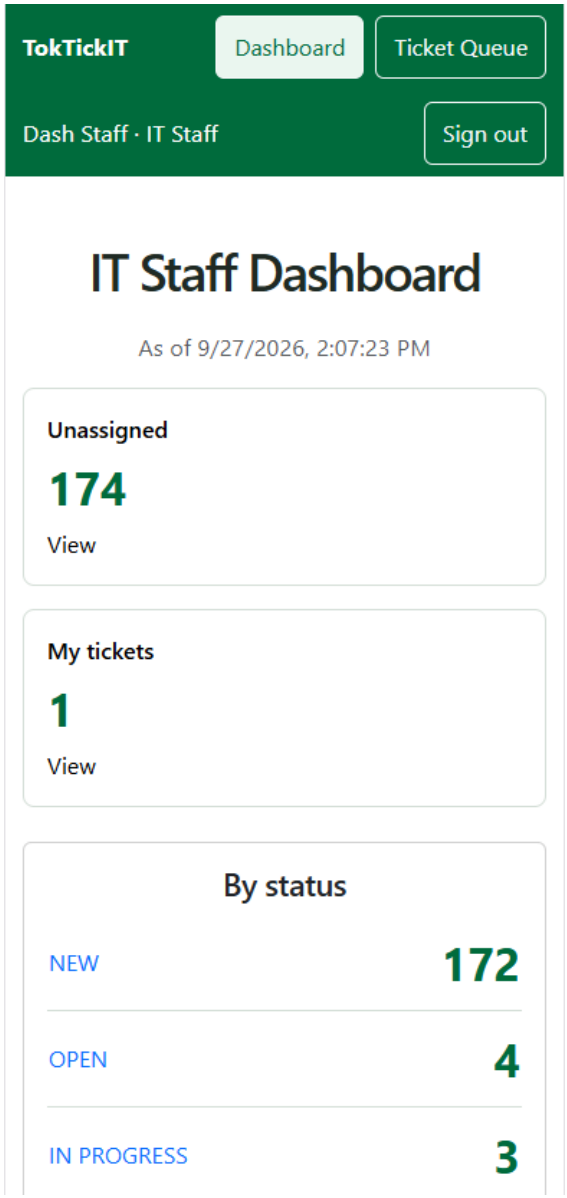
- Browser: `e2e/lab-04/dashboards.spec.ts` passed at 1440x1000, 820x1180 and 390x844 - a metric card drills into the filtered list, the filter banner clears, and a list row opens the ticket directly. The full Lab 4 suite (9 scenarios) and the full Lab 3 suite (15 scenarios) both still pass with the new Dashboard-first

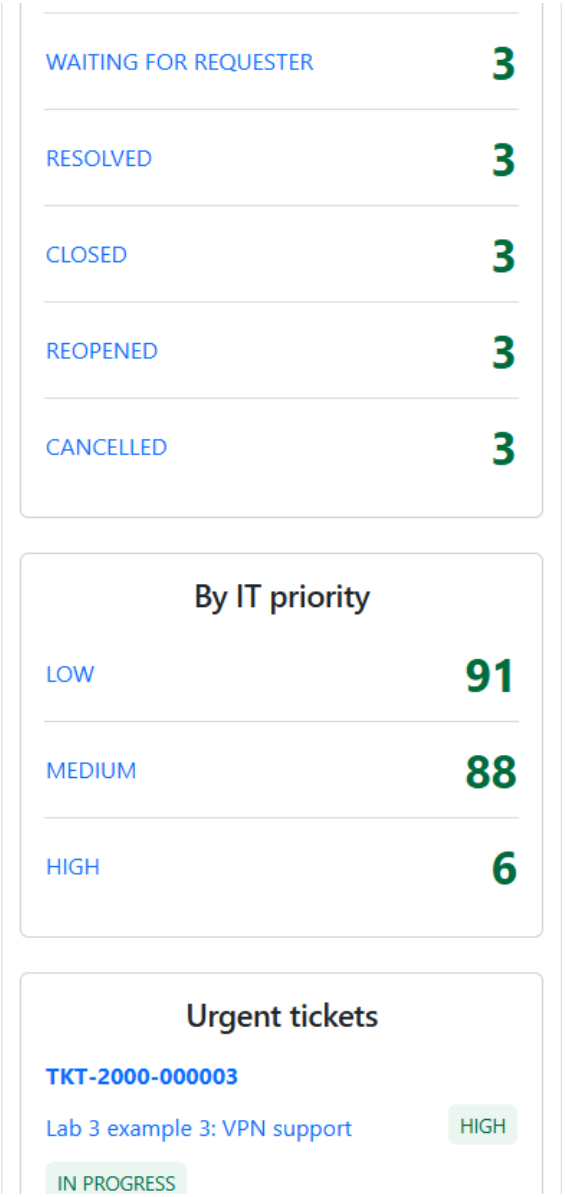
navigation. Screenshots inspected:
`artifacts/lab-04/screenshots/{staff-dashboard,requester-dashboard}/{desktop,tablet,mobile}.png`.

Peer review: approved by Kittakorn-P and merged as PR #60 (see reviewer.md). The remaining responsive, accessibility and regression rows in tests.md were completed in issue 6.

IT Staff dashboard - mobile

Original capture. Consecutive vertical excerpts; image content is unchanged.

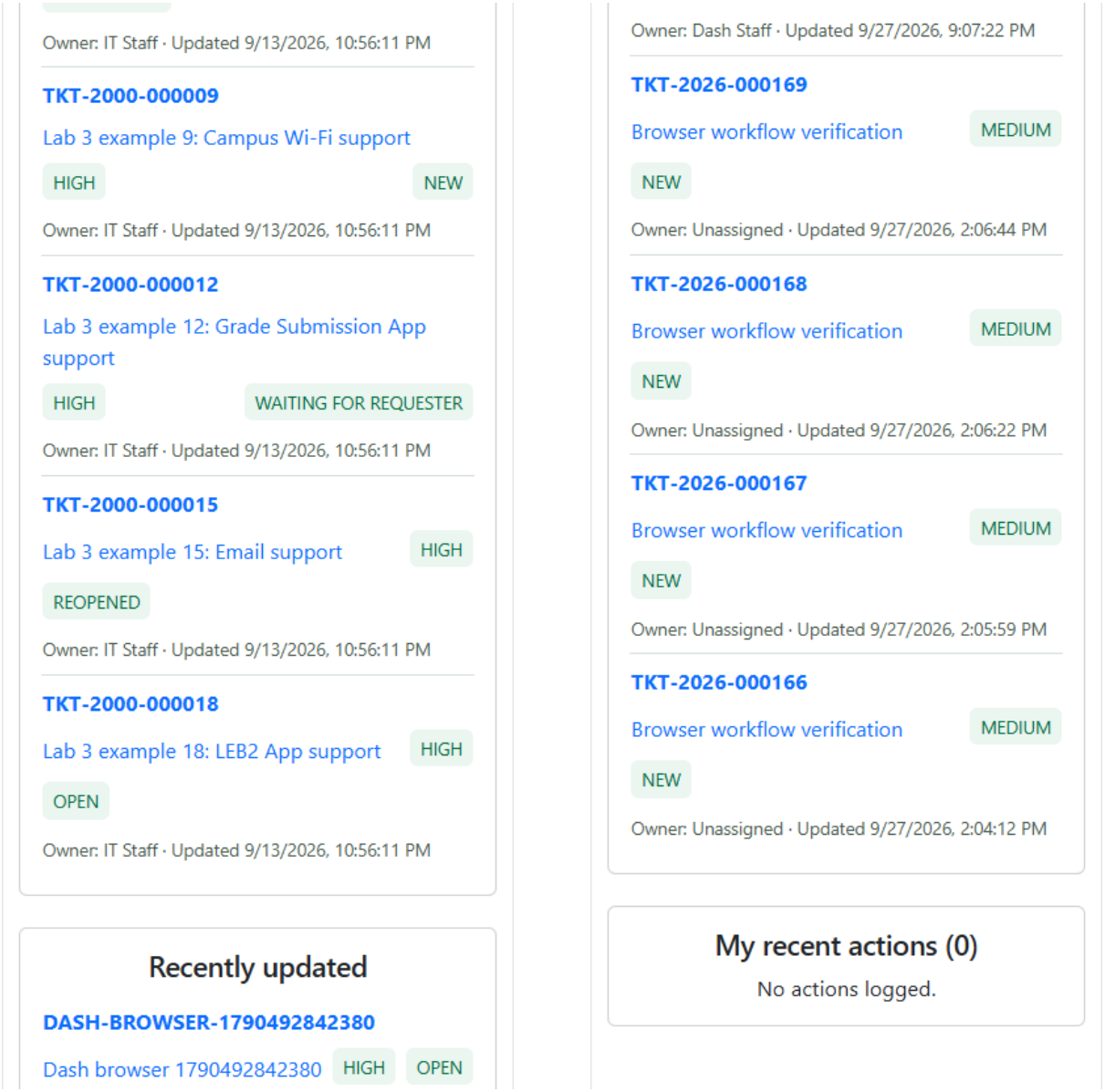




Open original PNG

IT Staff dashboard - mobile (continued)

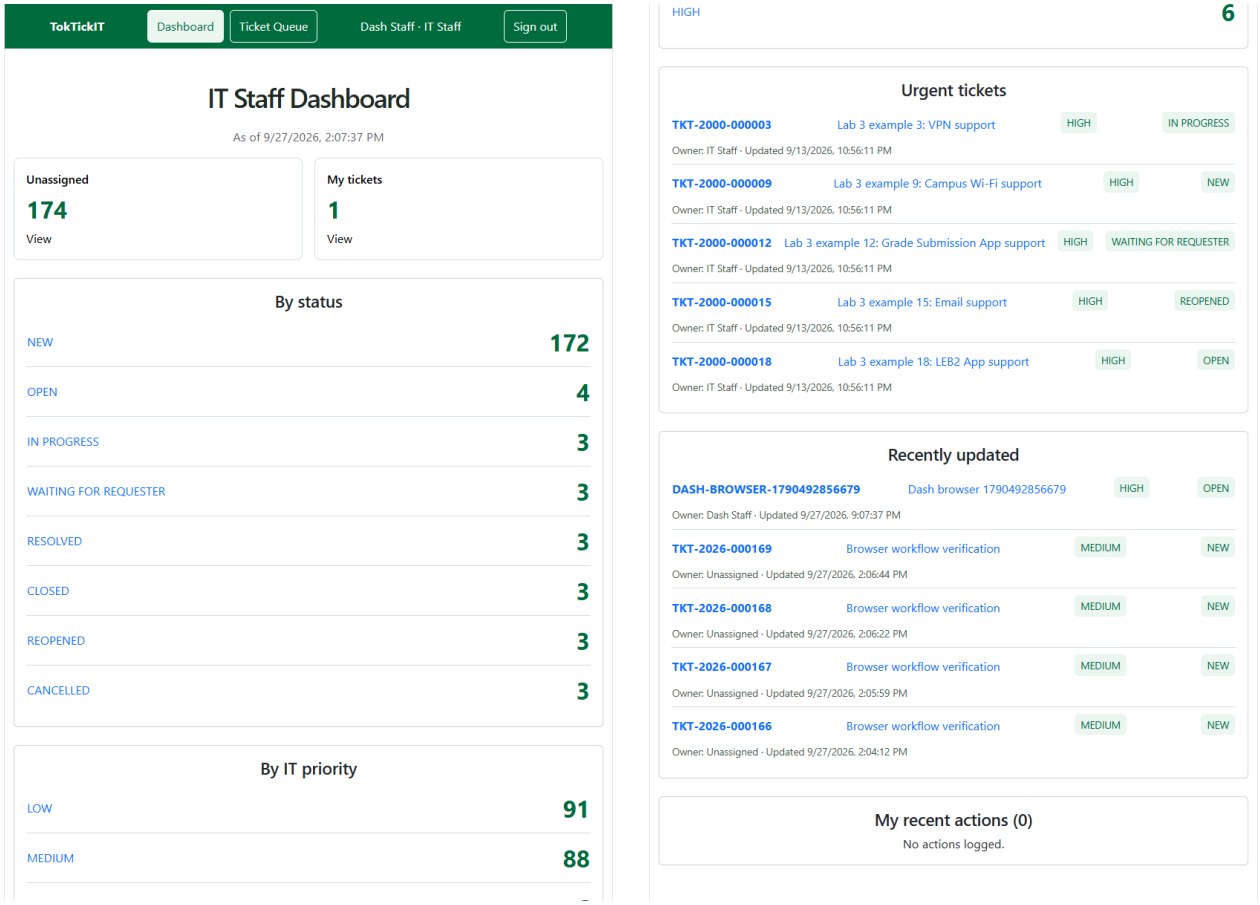
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

IT Staff dashboard - tablet

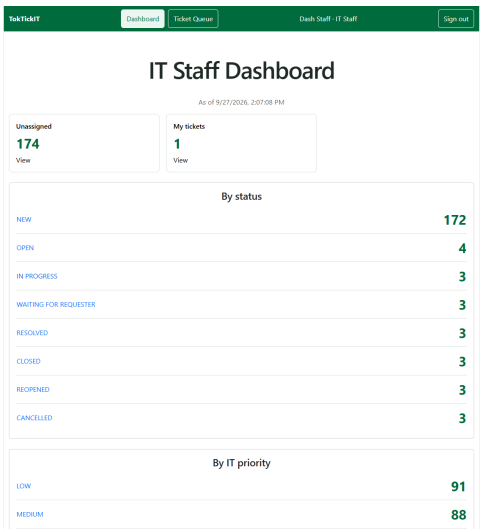
Original capture. Consecutive vertical excerpts; image content is unchanged.

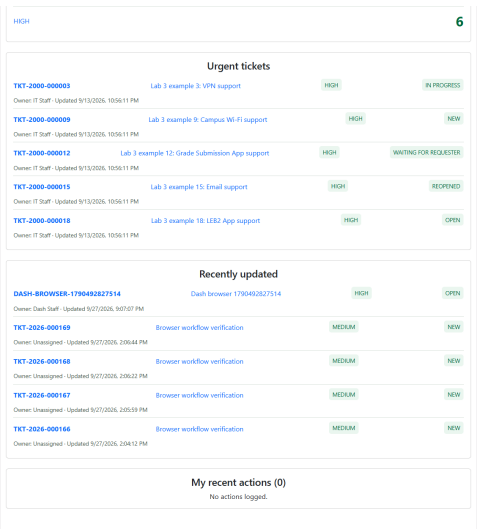


Open original PNG

IT Staff dashboard - desktop

Original capture. Consecutive vertical excerpts; image content is unchanged.





Open original PNG

Answer Part 6

Working Actions Taken UI

Real-browser scenarios list, add and edit actions, show different performers on one ticket, validate the follow-up rule, show the requester read-only view and lock a closed ticket. API tests cover inactive/forbidden roles, stale versions, duplicate request keys and safe failures (API-01..API-07).

Repository source

lab4 issue3 : Actions Taken UI

Depends on #50 (Actions Taken API, merged in PR #57). Adds the Actions Taken area to Ticket Detail for IT Staff, Administrators and Requesters.

What changed

- New `ActionsTaken` component, placed between the workflow card and Public Comments on Ticket Detail. It uses the Lab 4 API through `fetchActions`, `createAction` and `updateAction` in `client/src/api.ts`.

- List: date/time, performed by, description, result, follow-up (Yes with note / No) and attachment notes. A table on wide screens that stacks into labeled cards below 992px. Empty state, loading state and a safe load failure with "Reload actions".

- Create mode ("Add action") and edit mode (row "Edit"): description, result, follow-up checkbox, follow-up note (enabled and required only when follow-up is checked) and attachment notes. In edit mode performed by and the recorded date are shown read-only.

- Inline validation beside the fields with `aria-invalid`/`aria-describedby`, using the same limits as the server. The save button is disabled while saving. A failed save keeps the typed text and the same `requestKey`, so a retry cannot create a duplicate. A stale edit (409) shows the conflict message and a "Reload actions" button and keeps the draft.

- Requesters see the same list read-only. Closed and cancelled tickets hide Add/Edit for staff and explain why.

- Zen Green styles in `App.css`; buttons are at least 44px high on mobile.

Verification - 2026-09-24

- Client: 8 files, 35 tests passed (9 new in `client/tests/lab-04/ActionsTaken.test.tsx`; the Lab 3 `TicketActivity` test mock now also provides `fetchActions`). `tsc -b` and the Vite production build passed.

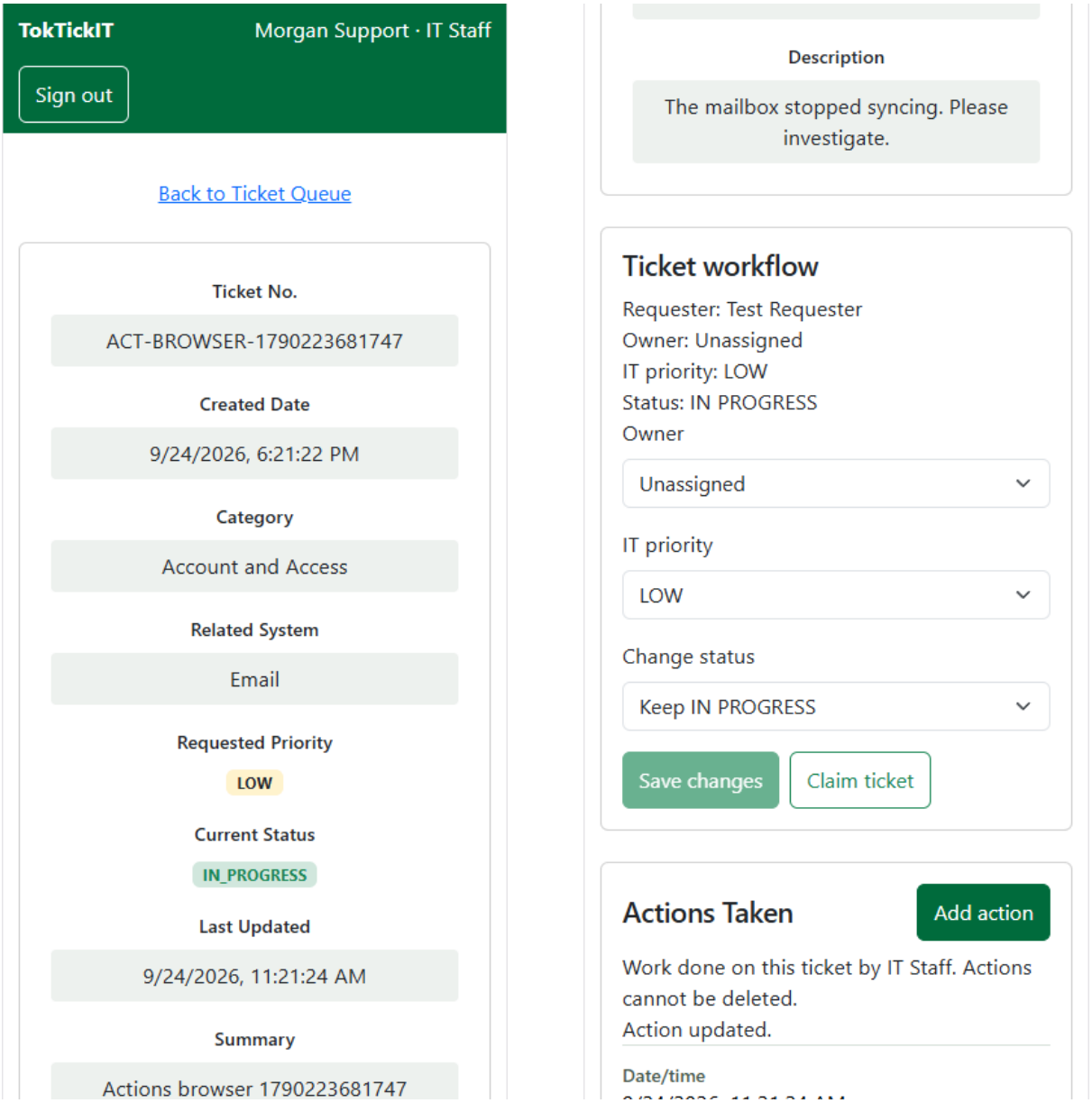
- Browser: `e2e/lab-04/actions-taken-flow.spec.ts` passed on desktop 1440x1000, tablet 820x1180 and mobile 390x844 against isolated `toktickit_lab4_test`. Flow: staff empty state, keyboard "Add action", validation, follow-up note rule, add, second staff member adds a different action and edits the first (performer stays the original author), requester read-only view, closed ticket read-only. The page-overflow assertion passed at all three widths. Screenshots inspected: `artifacts/lab-04/screenshots/actions-taken/{desktop,tablet,mobile}-{staff,requester}.png`.

Run browser tests: start the API with `PORT=4103` and `APP_ORIGIN=http://localhost:5183`, Vite with `VITE_API_URL=http://localhost:4103` on port 5183, both with `DATABASE_URL` pointing at `toktickit_lab4_test`, then from the repository root `npx playwright test --config playwright.lab4.config.ts`.

Peer review: approved by Kittakorn-P and merged as PR #58 (see reviewer.md). The status-control gate (issue 4), dashboards (issue 5) and final hardening (issue 6) are not part of this change.

Actions Taken, staff - mobile

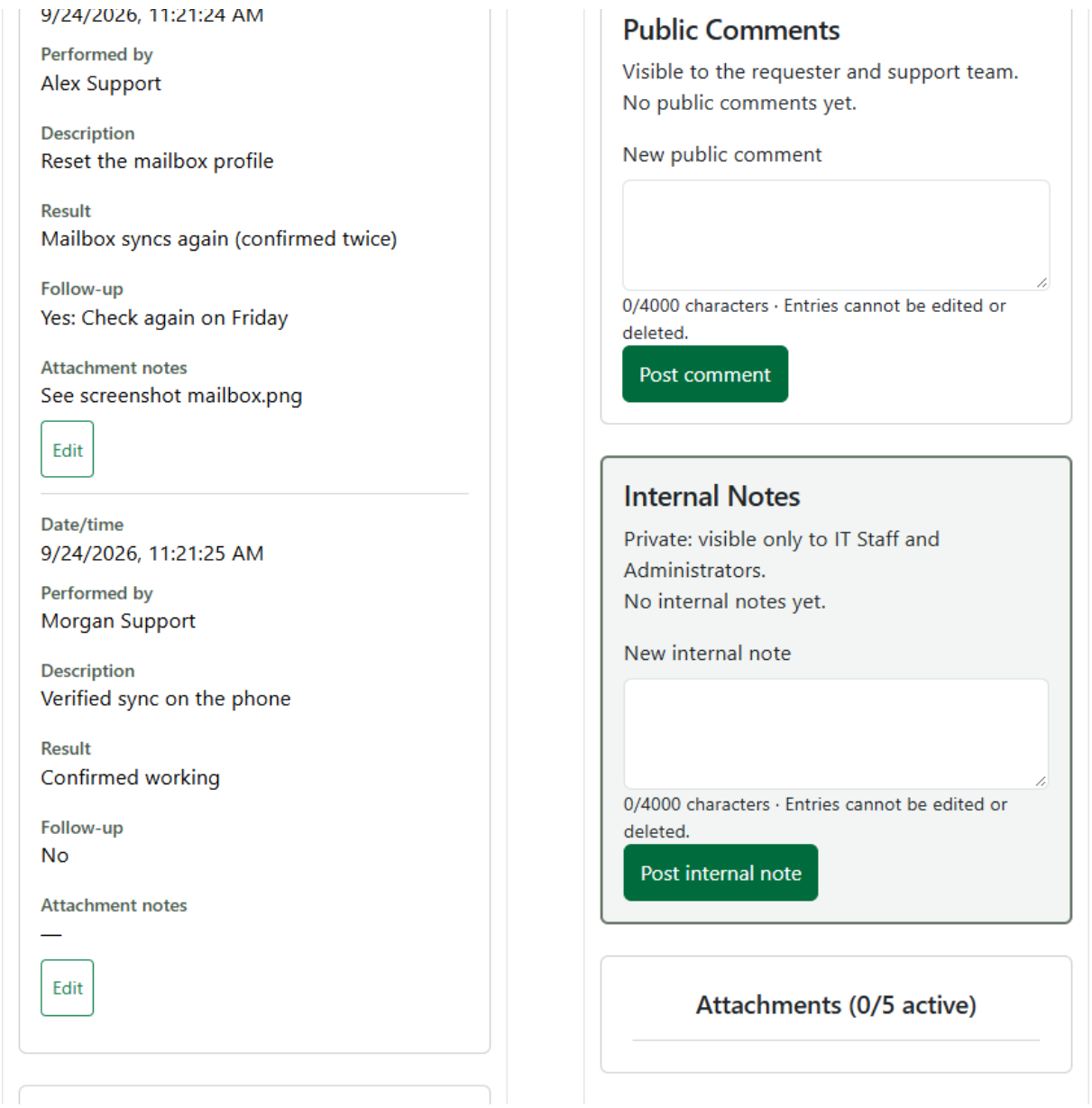
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

Actions Taken, staff - mobile (continued)

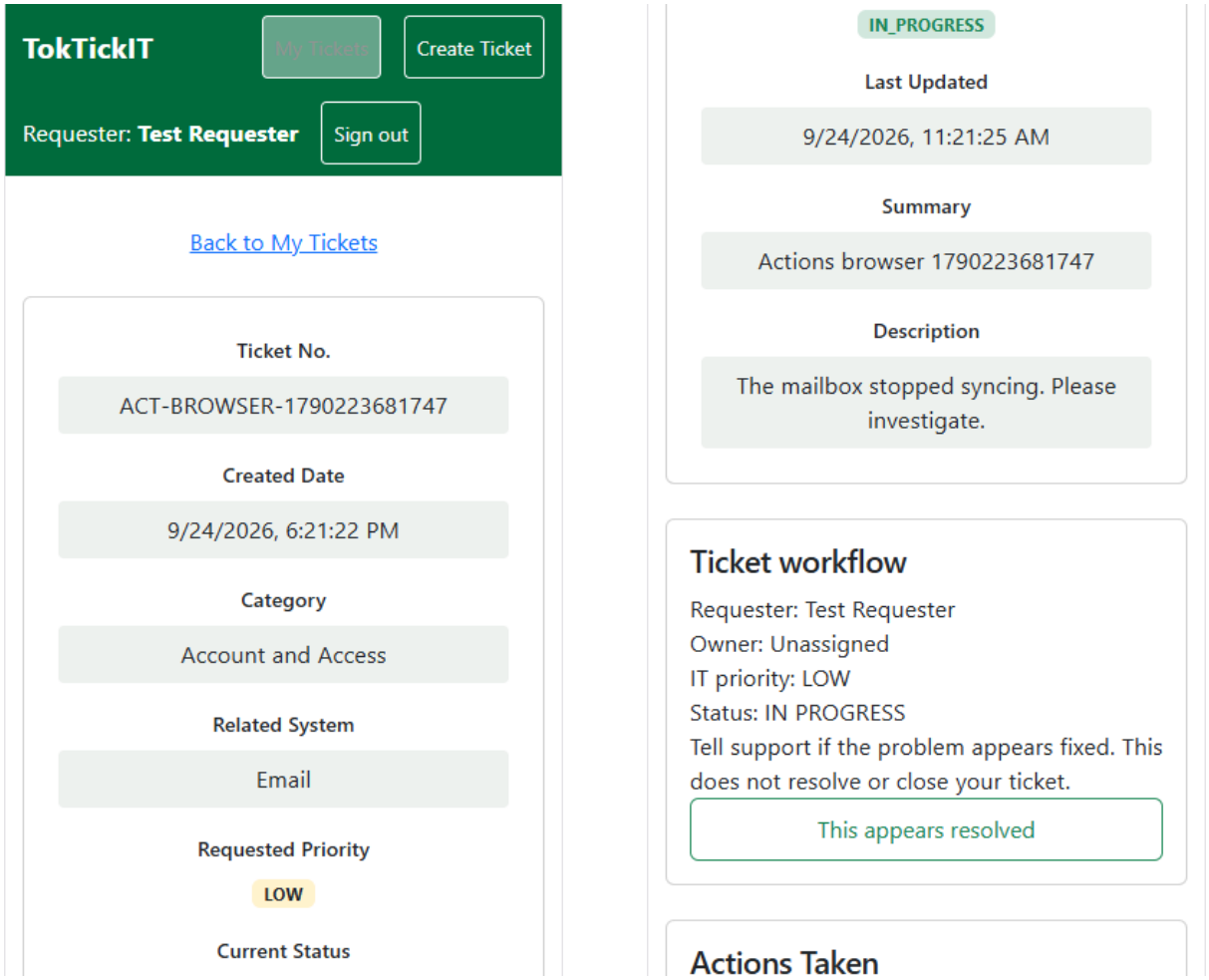
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

Actions Taken, requester read-only - mobile

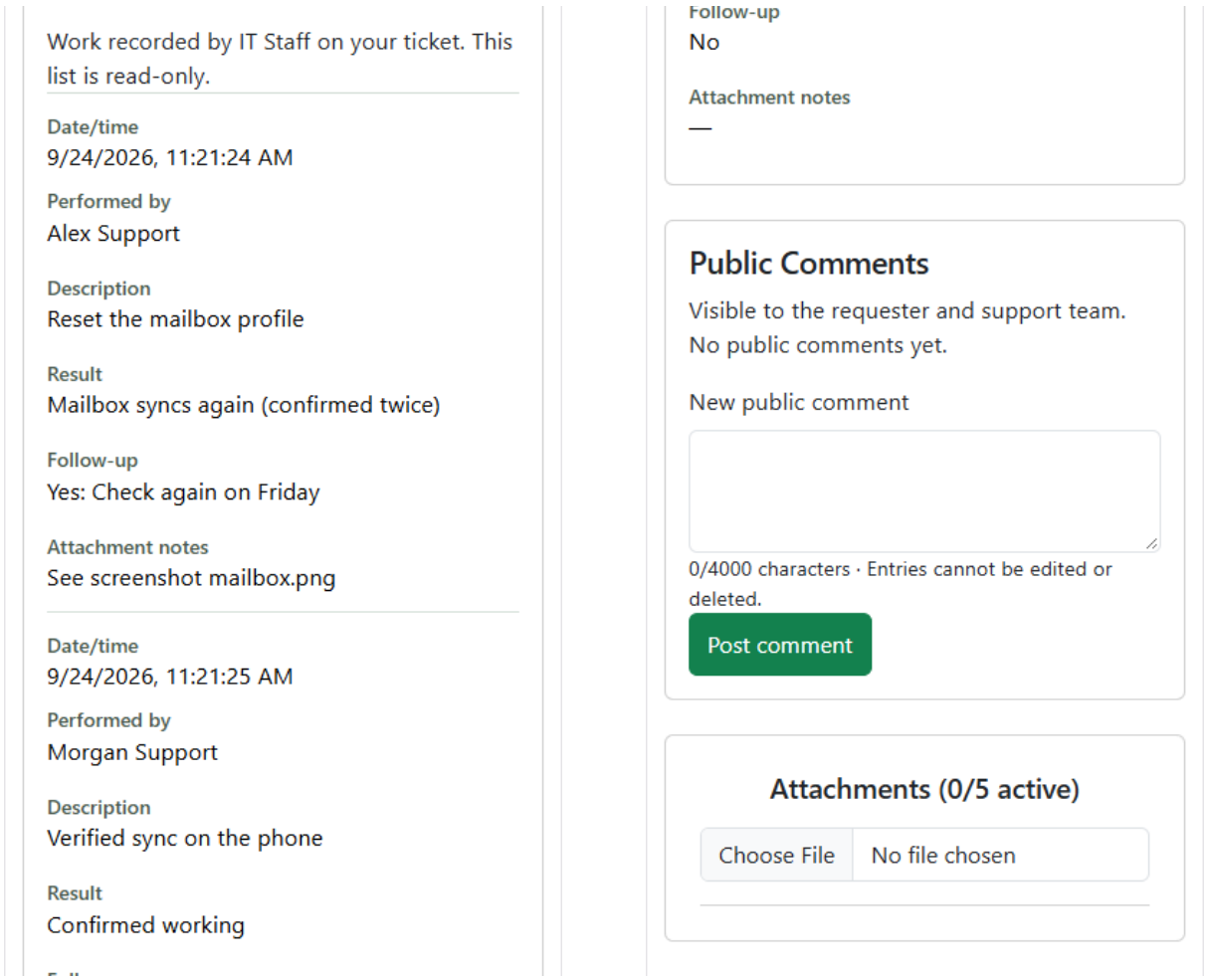
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

Actions Taken, requester read-only - mobile (continued)

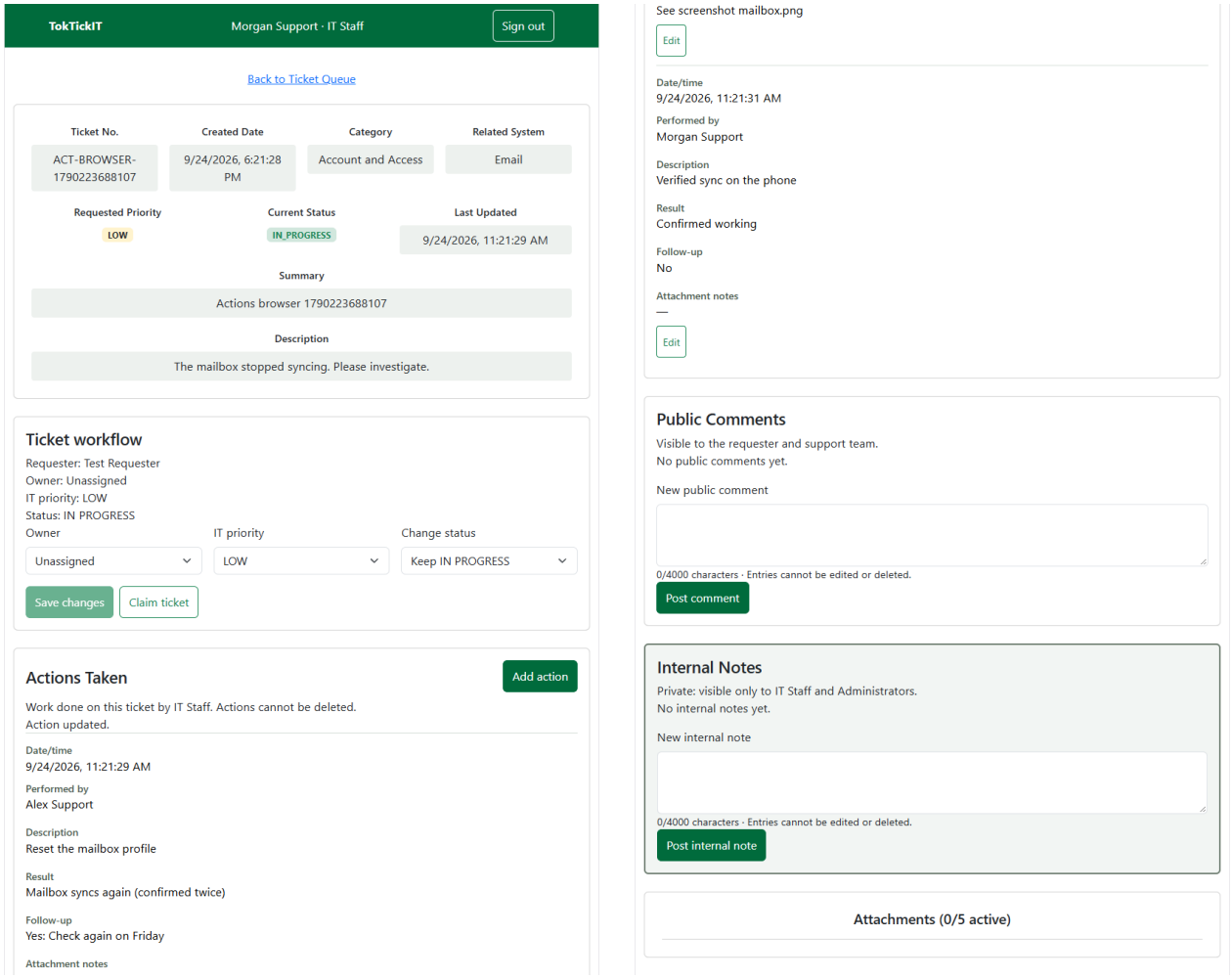
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

Actions Taken, staff - tablet

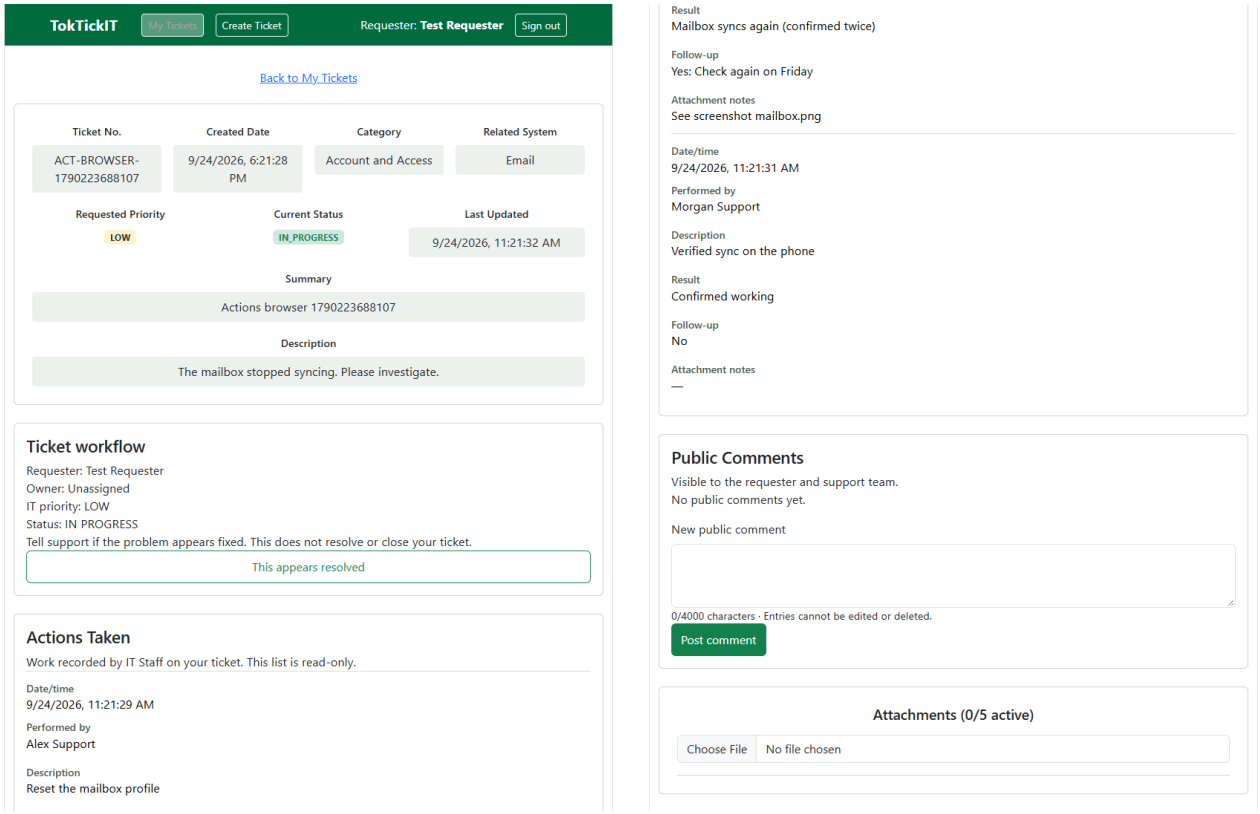
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

Actions Taken, requester read-only - tablet

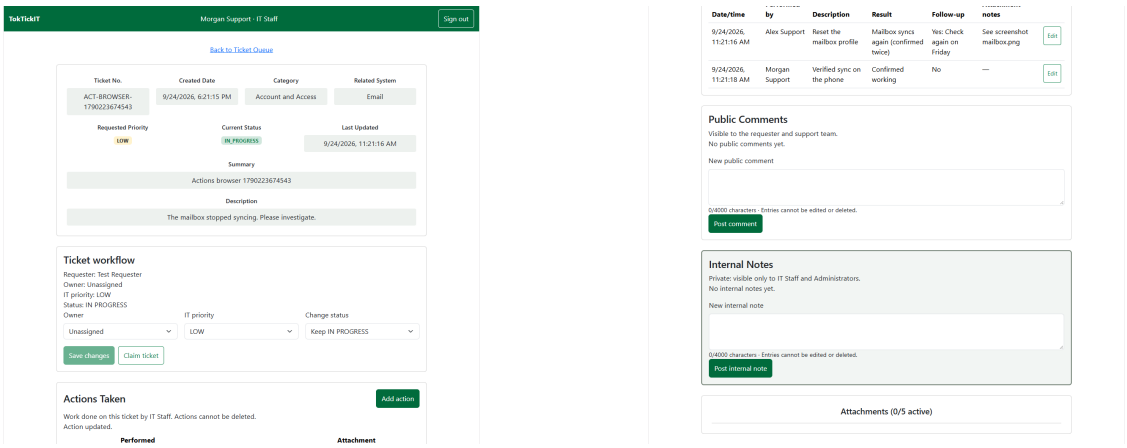
Original capture. Consecutive vertical excerpts; image content is unchanged.



Open original PNG

Actions Taken, staff - desktop

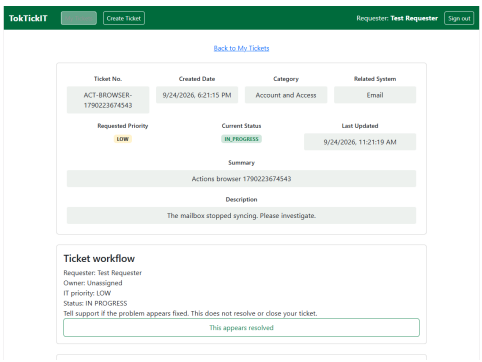
Original capture. Consecutive vertical excerpts; image content is unchanged.

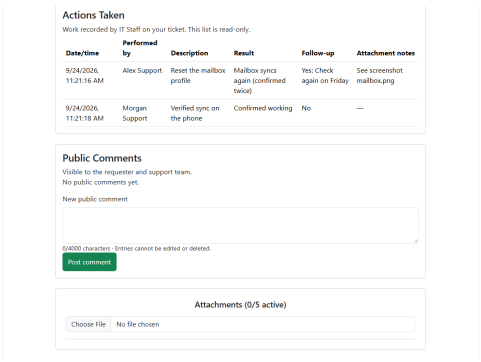


Open original PNG

Actions Taken, requester read-only - desktop

Original capture. Consecutive vertical excerpts; image content is unchanged.





Open original PNG

Answer Part 7

Working Ticket workflow

The transition matrix is tested for every from/to pair, the resolution gate returns ACTION_REQUIRED without an action, terminal statuses have no exits, and the requester signal never changes the status. Actions are ordered oldest first with an id tie-break and are never deleted (append-only).

Repository source

lab4 issue4 : Ticket workflow and resolution gate

Depends on #50 (Actions Taken API, PR #57) and #51 (Actions Taken UI, PR #58). Finalizes the ticket status workflow.

What changed

- **Resolution gate (BR-10):** moving a ticket to RESOLVED needs at least one Action Taken on that ticket, from any staff member. The check runs in the same transaction as the status update, under the ticket row lock, so a direct API call or a concurrent request cannot bypass it. Without an action the API returns 409 `ACTION_REQUIRED` and changes nothing. The gate applies only to RESOLVED; other moves, edits and cancellation work without actions.

- **Final transition matrix:** unchanged destinations from Lab 3 (specification 5.2), still enforced by the backend for IT Staff only; requesters and administrators get 403. CLOSED and CANCELLED have no exits; RESOLVED, CLOSED and CANCELLED still need confirmation; stale versions return 409 `TICKET_CONFLICT`. Error codes are now carried per conflict instead of always `TICKET_CONFLICT`.

- **Requester signal:** "This appears resolved" stays advisory. It never changes the status and does not skip the gate.

- **UI:** the status control lists only permitted next statuses (none beyond "Keep" for closed/cancelled). Choosing Resolved on a ticket with no actions shows "Log an action first" and disables Save; the server error is still shown if it rejects. `ActionsTaken` reports its count to the workflow card. A successful save reloads the ticket, so the summary status badge and workflow card show the new status.

- **Lab 3 tests updated for the new rule:** the Lab 3 server workflow fixture and client `TicketActivity` mock now give the ticket one action, and the Lab 3 workflow browser test logs an action before resolving. The Lab 3 e2e database guards accept the Lab 4 test database and `DATABASE_URL`.

Verification - 2026-09-27

- Backend: 17 suites, 251 tests passed against isolated `toktickit_lab4_test`; every from/to status pair is tested in `server/tests/lab-04/ticket-workflow.api.test.ts`, plus the gate, roles, unconfirmed/stale updates, terminal statuses, advisory signal and concurrent resolve. TypeScript build passed.

- Client: 9 files, 47 tests passed (12 new in `client/tests/lab-04/TicketWorkflow.test.tsx`); `tsc -b` and Vite build passed.

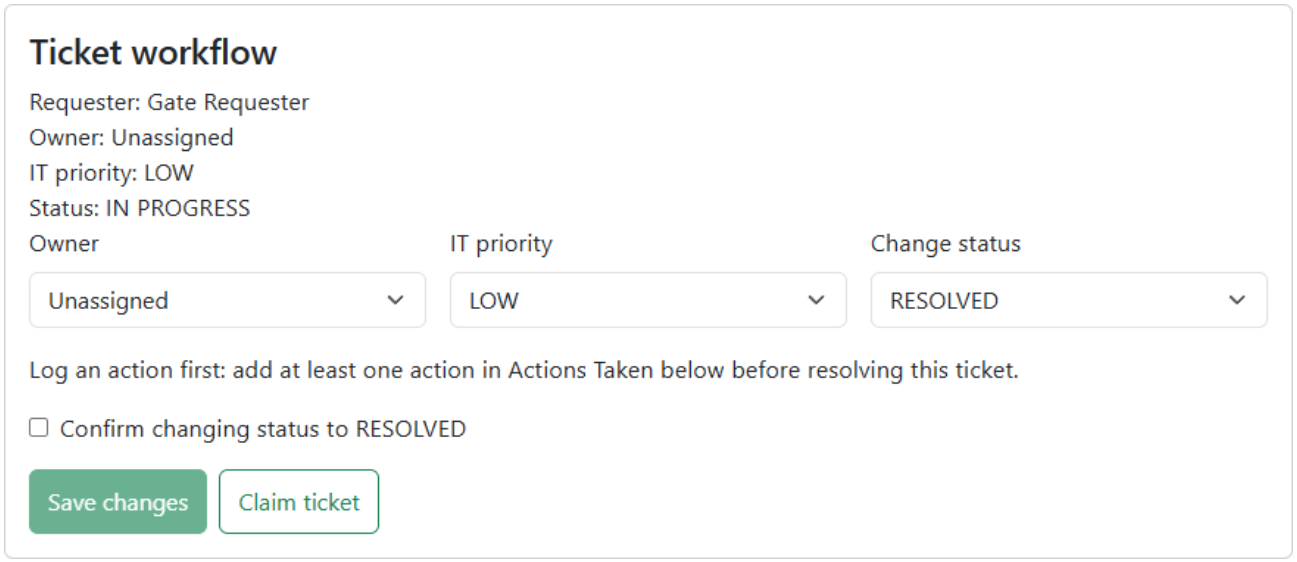
- Browser: `e2e/lab-04/ticket-resolution.spec.ts` passed at 1440x1000, 820x1180 and 390x844; Lab 4 actions spec still passes; the full Lab 3 browser suite passes (15 scenarios) after the gate change. Screenshots inspected: `artifacts/lab-04/screenshots/ticket-workflow/{desktop,tablet,mobile}-{gate-blocked,resolved}.png`.

Run the browser tests as in issue-03.md, then `npx playwright test --config playwright.lab4.config.ts` and `--config playwright.lab3.config.ts`.

Peer review: approved by Kittakorn-P and merged as PR #59 (see reviewer.md). Dashboards (issue 5) and final hardening (issue 6) are not part of this change.

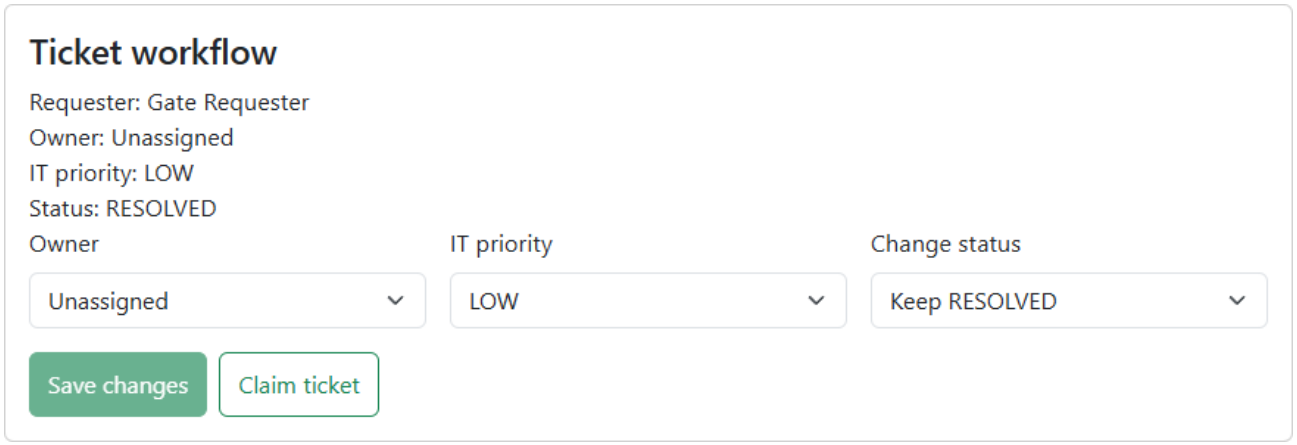
Ticket workflow - desktop

Resolved blocked until an action is logged



Full-resolution screenshot

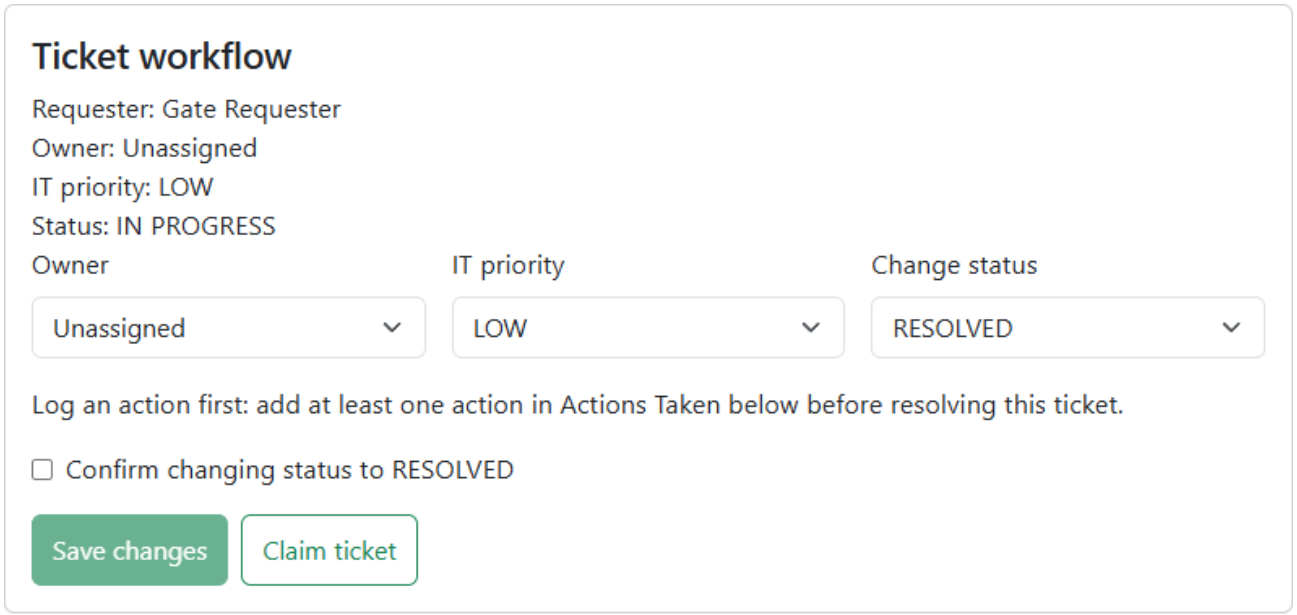
After an action is logged: Resolved, next options Closed or Reopened



Full-resolution screenshot

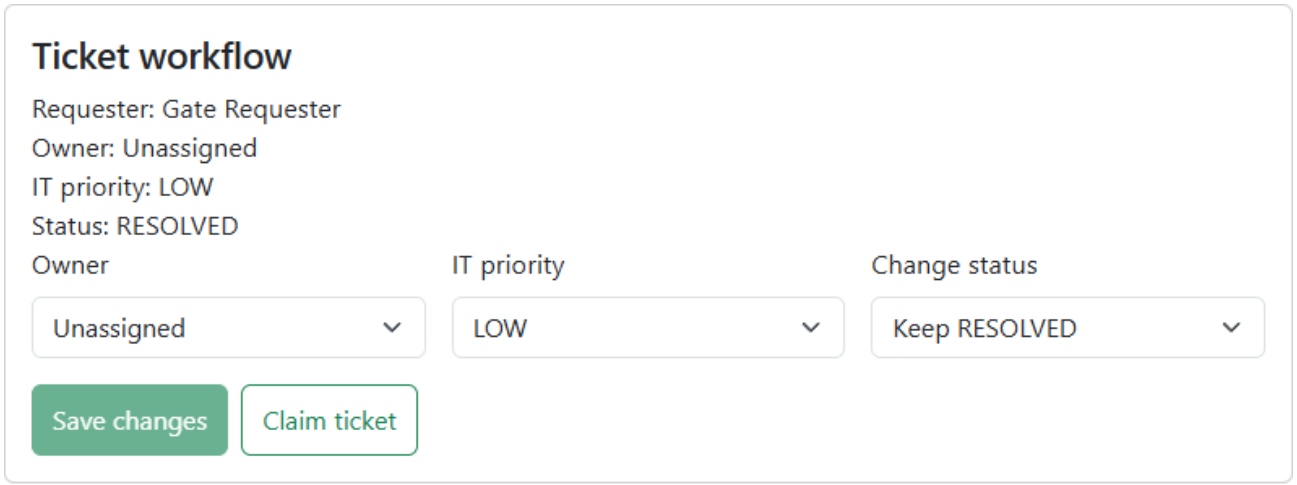
Ticket workflow - tablet

Resolved blocked until an action is logged



Full-resolution screenshot

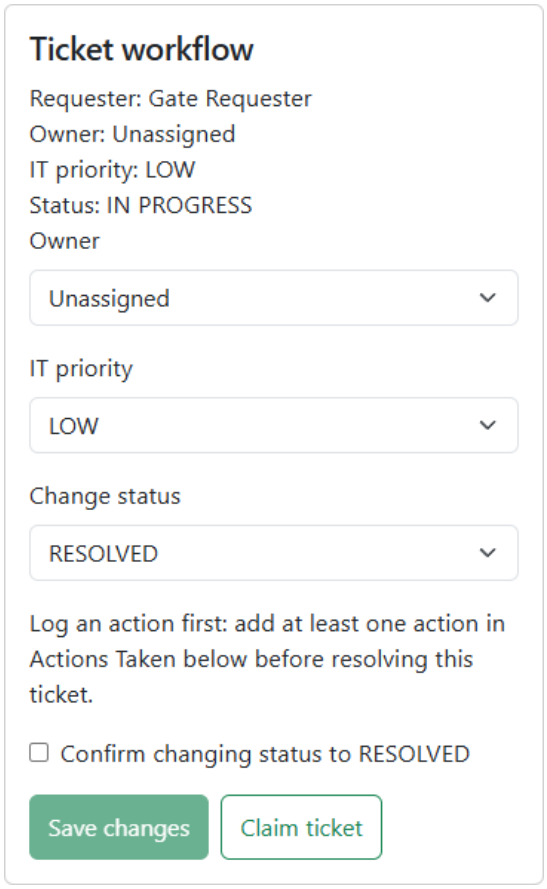
After an action is logged: Resolved, next options Closed or Reopened



Full-resolution screenshot

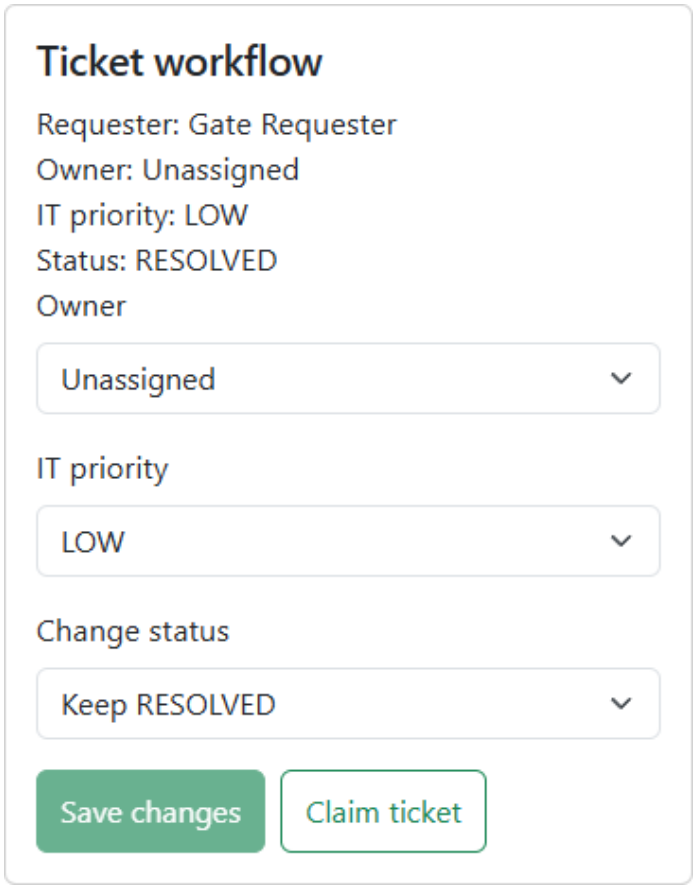
Ticket workflow - mobile

Resolved blocked until an action is logged



Full-resolution screenshot

After an action is logged: Resolved, next options Closed or Reopened



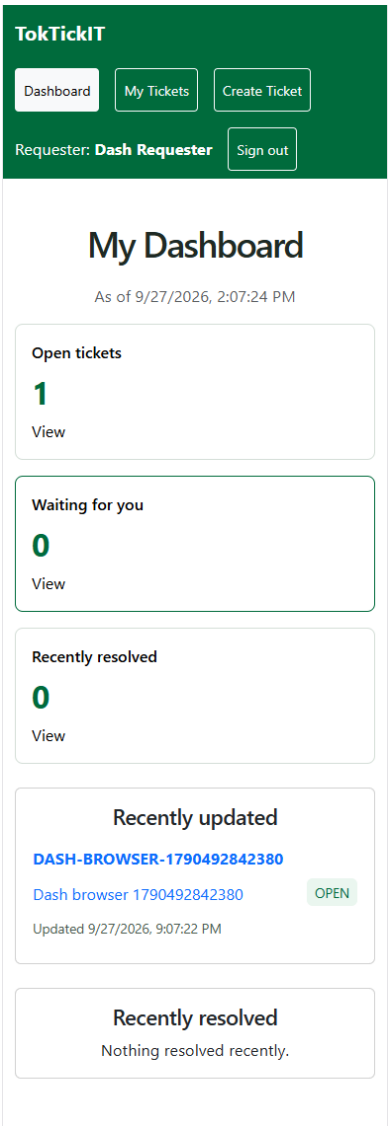
Full-resolution screenshot

Answer Part 8

Requester dashboard and final regression

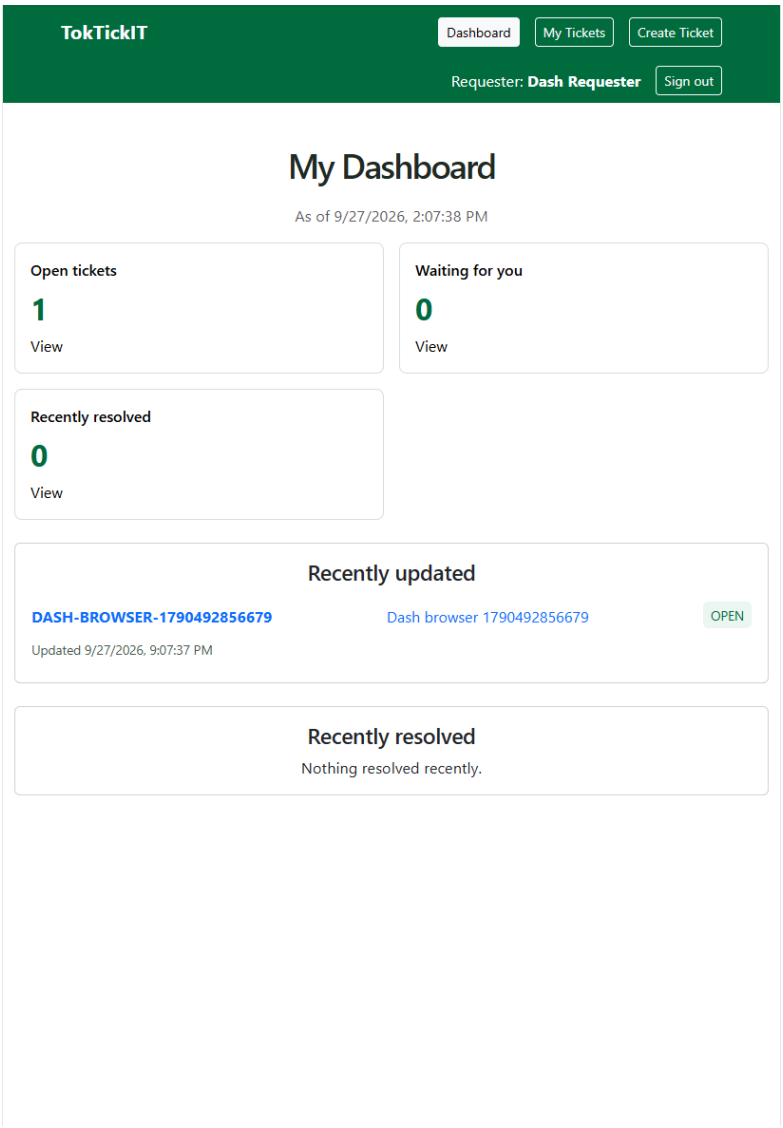
The Requester dashboard returns only the caller's own metrics (API-10), empty accounts show zeros (API-13), and counts equal the drill-down list. Regression evidence: the full Lab 3 browser suite (15 scenarios covering login, My Tickets, ticket detail, attachments, public comments, staff functions, internal notes and administrator user management) and all Lab 1-3 server and client tests pass on main (Part 3).

Requester dashboard - mobile



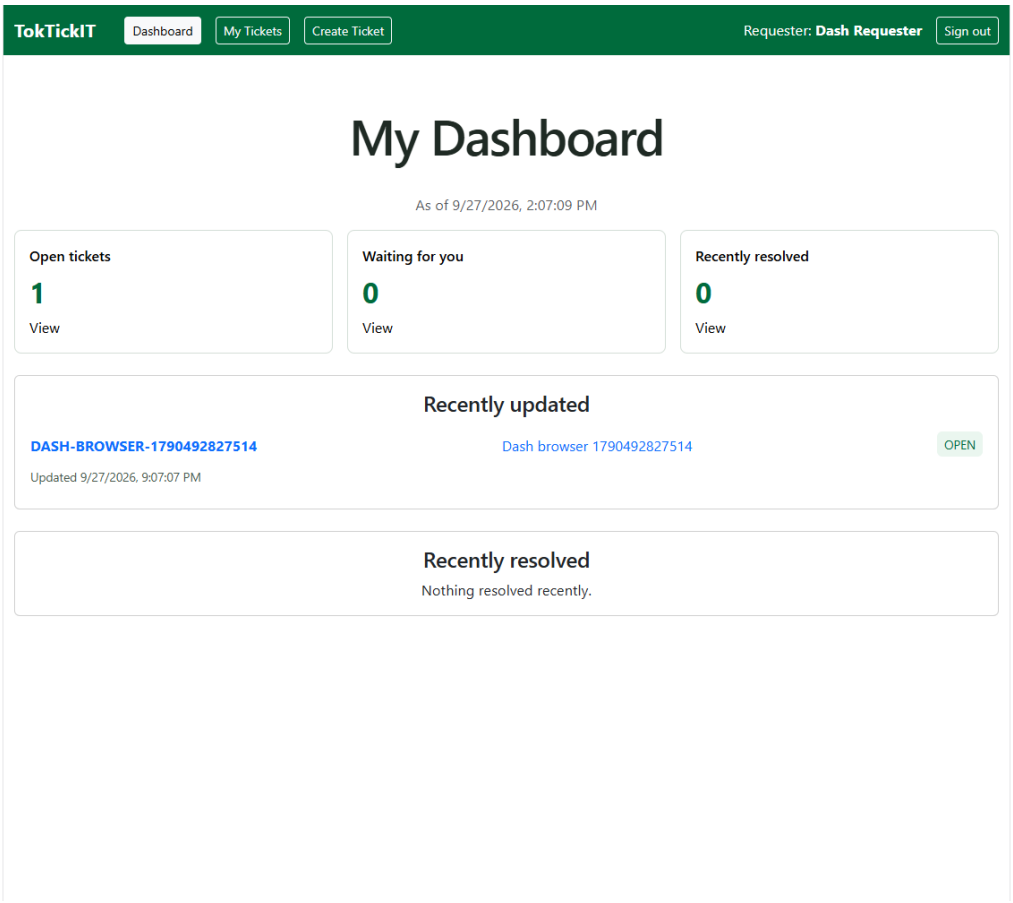
Full-resolution screenshot

Requester dashboard - tablet



Full-resolution screenshot

Requester dashboard - desktop



Full-resolution screenshot

Answer Part 9

Zen Green UI, responsive behavior, accessibility and final polish

Repository source

Lab 4 UI specification

Reuses Lab 2/3 Zen Green tokens (primary #006B3C, secondary #0B7A46, pale #EAF6EF, page #F5F7F6, text #1F2A24, read-only #EDF1EE, error #B3261E), shared Badge, Feedback, forms, cards and tables. 16px base type, 8px unit, 44px min touch target on mobile, visible focus ring, text (not color-only) status cues.

Screen / mode	Structure and behavior
Shell	Adds **Dashboard** first in nav for every role (Requester: Dashboard, My Tickets, Create; Staff: Dashboard, Queue; Admin: Dashboard, Queue, Users). Active page has `aria-current="page"` and a distinct style. Role home after login = Dashboard. Obsolete or duplicate controls from earlier labs are removed.
Staff dashboard	Heading + "as of" time. Metric cards: Unassigned, My tickets (each a link with label, large value, "View" text). "By status" and "By IT priority" lists, each row a link with count. Lists: Urgent, Recently updated, My recent actions (rows link to Ticket Detail). Admin: extra "User accounts" card. Desktop 3 columns, tablet 2, mobile 1.
Requester dashboard	Cards: Open tickets, Waiting for you (text badge "Needs your attention" when >0), Recently resolved; lists Recently updated and Recently resolved. Links to My Tickets with filters. Does not replicate the full list.
Dashboard states	Loading text `role=status`; failure `role=alert` with Retry; 403 shows a plain "not available for your role" message with a home link; zero data shows "0" cards and per-list empty text.
Actions Taken panel (staff detail)	Table on >=992px, stacked cards below. Columns: Date/time, Performed by, Description, Result, Follow-up (Yes/No + note), Attachment notes. "Add action" opens **create mode** (description, result, follow-up checkbox, follow-up note enabled and required when checked, attachment notes). Each row has "Edit" -> **edit mode** (same form, Save/Cancel). Performed-by and date are read-only.
Actions Taken (requester detail)	Same list, read-only, no buttons.
Actions form feedback	Inline errors beside fields (`aria-invalid`, `aria-describedby`); Save disabled while busy; double-click cannot duplicate (requestKey); failure keeps entered text; conflict message "This action changed, reload"; terminal Tickets hide Add/Edit and show "Ticket is closed". Success uses `role=status` and refreshes the list and ticket summary.
Ticket workflow	Status select lists only allowed next statuses; RESOLVED with zero Actions shows the hint "Log an action first" and, if bypassed, the API error `ACTION_REQUIRED`. Confirmation dialog (focus trapped, Esc closes, focus returns) for RESOLVED/CLOSED/CANCELLED. After success the summary status badge and version refresh.

Responsive: >=992px multi-column; 768-991px two columns/cards; <768px stacked, wrapped buttons and long text, no horizontal page scroll. Container max 1200px.

Visual evidence at 1440x1000, 820x1180 and 390x844 in `artifacts/lab-04/screenshots/{staff-dashboard,requester-dashboard,actions-taken}/`.

Visual and accessibility checklist (tick only after actual inspection)

- [ ] Zen Green tokens and spacing consistent across all new screens

- [ ] Active nav indicated; role nav correct

- [ ] Metric cards: label, value, keyboard-operable drill-down link

- [ ] Editable fields white, read-only fields grey; performed-by/date read-only

- [ ] Validation placed beside fields and announced

- [ ] Keyboard focus visible on all controls; dialog focus trapped and restored

- [ ] Status/priority/private-vs-shared cues are text, not color only

- [ ] No clipping, overlap or horizontal overflow at 390/820/1440 widths

- [ ] No console errors, placeholder text or unfinished controls

Evidence for the checklist above: the accessibility browser test (accessibility.spec.ts, RESP-01 and A11Y-01 in Part 3) checks real keyboard navigation with a visible focus ring, one main landmark and one h1 and one active nav item per screen, and no console errors or failing API calls for all three roles at 1440x1000, 820x1180 and 390x844. Issue 6 restored a focus ring Bootstrap was hiding and moves focus to the first invalid field. Desktop, tablet and mobile screenshots of every major Lab 4 screen are in Parts 5-8.

Repository source

lab4 issue6 : Final hardening and regression

Depends on #51, #52 and #53 (all merged). Final polish pass over the complete application plus the full Lab 1-4 regression.

What changed

- **Leftovers removed:** the obsolete Lab 2 development requester-selection screen (`RequesterSelection.tsx`) and the dead `fetchRequesters()` call to the removed `/api/requesters` endpoint. A search found no `console.*` calls, TODO markers or placeholder text in the client source.

- **Keyboard focus:** Bootstrap removes the outline on `.btn:focus-visible`, which left nav and action buttons with a faint ring. Added a clear 3px outline for buttons and dashboard cards (white on the green navigation bar).

- **Validation focus:** on a failed Actions Taken save, keyboard focus now moves to the first invalid field (with `aria-invalid` and `aria-describedby` already in place).

- **Already covered earlier in Lab 4 (verified again here):** duplicate-submit protection (disabled while saving, same `requestKey` on retry), drafts kept after recoverable failures, conflict/forbidden/not-found/empty/loading feedback, stale-update 409s.

- **README** now documents Lab 4 setup, migration and rollback, seed, test database, the integrated verification command and a demonstration path; `server/.env.example` names the Lab 4 test database.

- `scripts/verify-lab4.cjs` runs backend and client tests, type checks, build, the migration check and the Lab 4 and Lab 3 browser suites, recording each command, exit code and log in `artifacts/lab-04/verification/`.

Verification

New accessibility browser test `e2e/lab-04/accessibility.spec.ts` (all three roles, three viewports): keyboard-only navigation with real Tab focus, visible focus ring, one `main`, one `h1` and one active nav item per screen, and no console errors or failing API calls across dashboards, queue, My Tickets, Create Ticket,

ticket detail and Users. Its first run found the weak focus ring above, which is now fixed. Full integrated run (`node scripts/verify-lab4.cjs`, 2026-09-28): server 19 suites / 261 tests, client 11 files / 59 tests, type checks, build and migration check pass; Lab 4 browser suite 12 scenarios and Lab 3 regression 15 scenarios pass at all three viewports. Logs and `report.json` are in `artifacts/lab-04/verification/` (the report records the base commit because it ran on the uncommitted issue 6 tree; issue 7 repeats it on final main). `tests.md` now shows every planned test as Pass.

Peer review: approved by Kittakorn-P and merged as PR #61 (see reviewer.md). Release integration into `main`, `ai-use.md` and the submission PDF are issue 7.